

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/switched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4

```

```
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	7988
Nombre de lignes df clean	7988

Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5591
Taille dataset test	2396
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_24h_mean, r_NH3_st_24h_std, r_NH3_st_24h_rms, r_NH3_st_24h_median, r_NH3_st_24h_min, r_NH3_st_24h_max, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_temperature_st_24h_mean, r_temperature_st_24h_std, r_temperature_st_24h_rms, r_temperature_st_24h_median, r_temperature_st_24h_min, r_temperature_st_24h_max, r_temperature_st_24h_diff_last, r_temperature_st_24h_slope, r_umidity_st_24h_mean, r_umidity_st_24h_std, r_umidity_st_24h_rms, r_umidity_st_24h_median, r_umidity_st_24h_min, r_umidity_st_24h_max, r_umidity_st_24h_diff_last, r_umidity_st_24h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

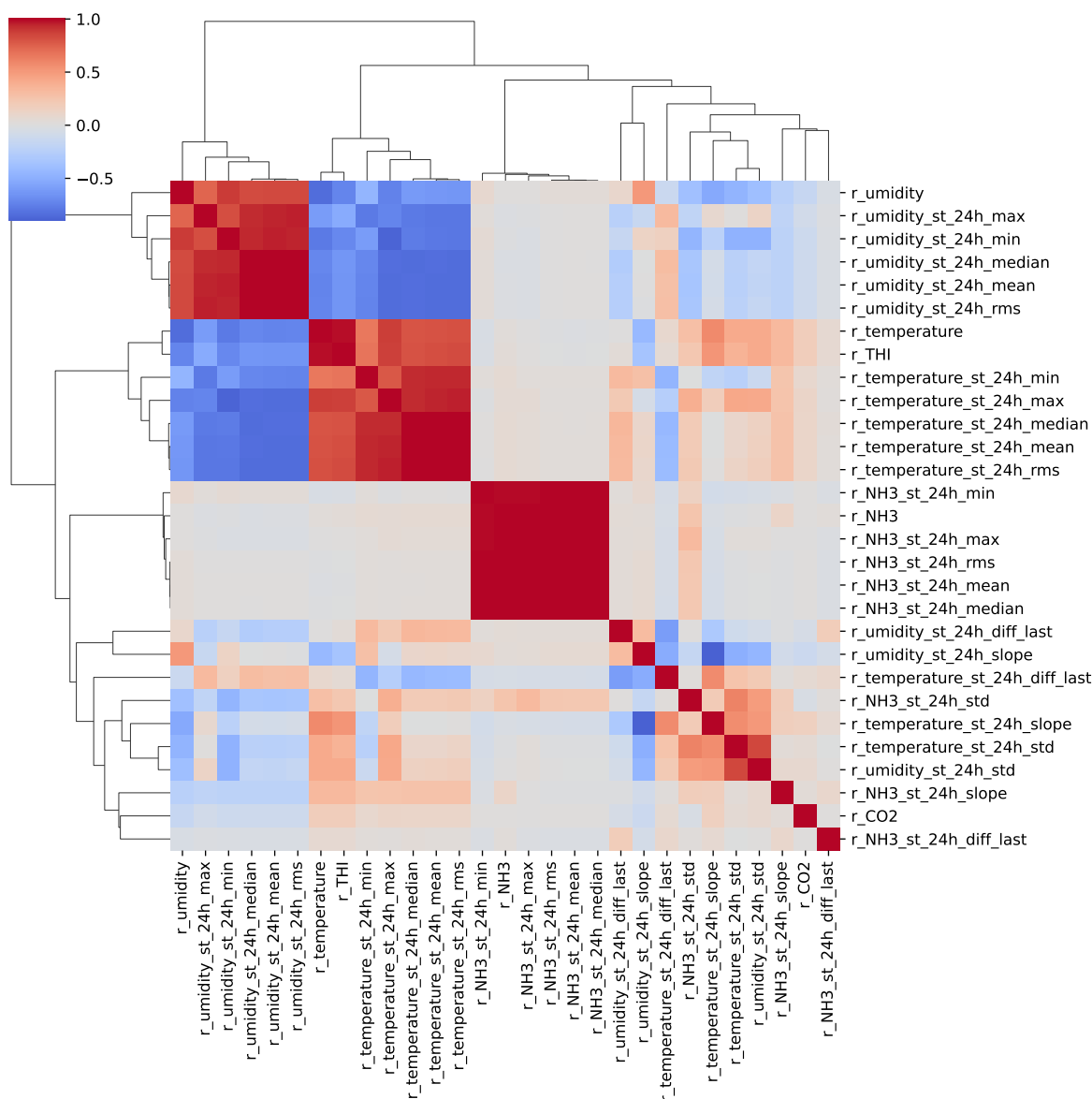
cluster 1: r_NH3_st_24h_median, r_NH3_st_24h_rms, r_NH3_st_24h_mean, r_NH3_st_24h_min, r_NH3, r_NH3_st_24h_max ; **cluster 2:** r_umidity_st_24h_max, r_temperature_st_24h_max, r_umidity_st_24h_min, r_umidity, r_temperature_st_24h_min, r_THI, r_umidity_st_24h_rms, r_temperature_st_24h_mean, r_temperature_st_24h_median, r_temperature_st_24h_rms, r_umidity_st_24h_mean, r_temperature, r_umidity_st_24h_median ; **cluster 3:** r_CO2 ; **cluster 4:** r_NH3_st_24h_std ; **cluster 5:** r_NH3_st_24h_diff_last ; **cluster 6:** r_NH3_st_24h_slope ; **cluster 7:** r_umidity_st_24h_std, r_temperature_st_24h_std ; **cluster 8:** r_temperature_st_24h_diff_last

; **cluster 9:** r_umidity_st_24h_slope, r_temperature_st_24h_slope ; **cluster 10:**
r_umidity_st_24h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chosen features_X: r_NH3_st_24h_median, r_umidity_st_24h_max, r_CO2,
r_NH3_st_24h_std, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_umid-
ity_st_24h_std, r_temperature_st_24h_diff_last, r_umidity_st_24h_slope, r_umid-
ity_st_24h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3_st_24h_median, r_umidity_st_24h_max, r_CO2, r_NH3_st_24h_std, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_umidity_st_24h_std, r_temperature_st_24h_diff_last, r_umidity_st_24h_slope, r_umidity_st_24h_diff_last

Len(Features_X): 10

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

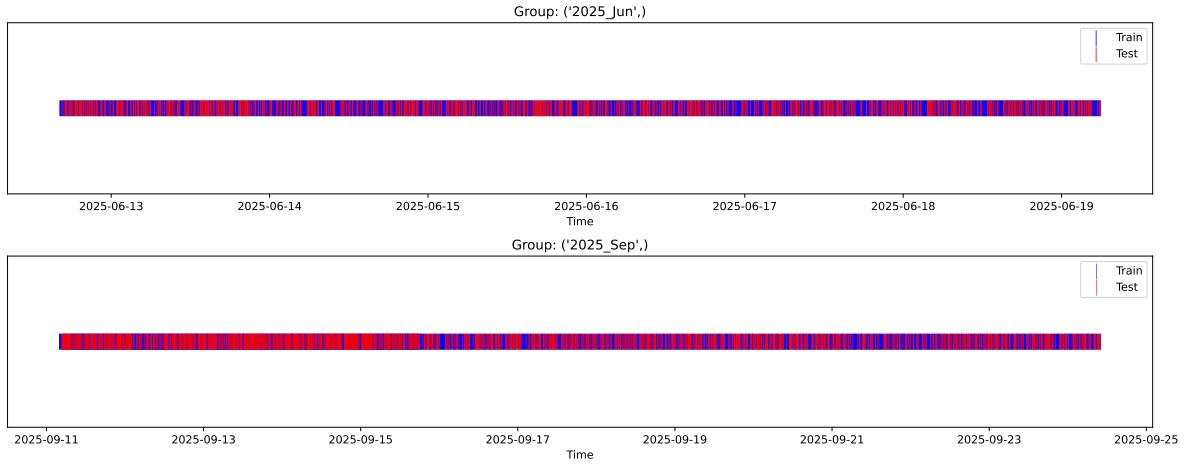
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

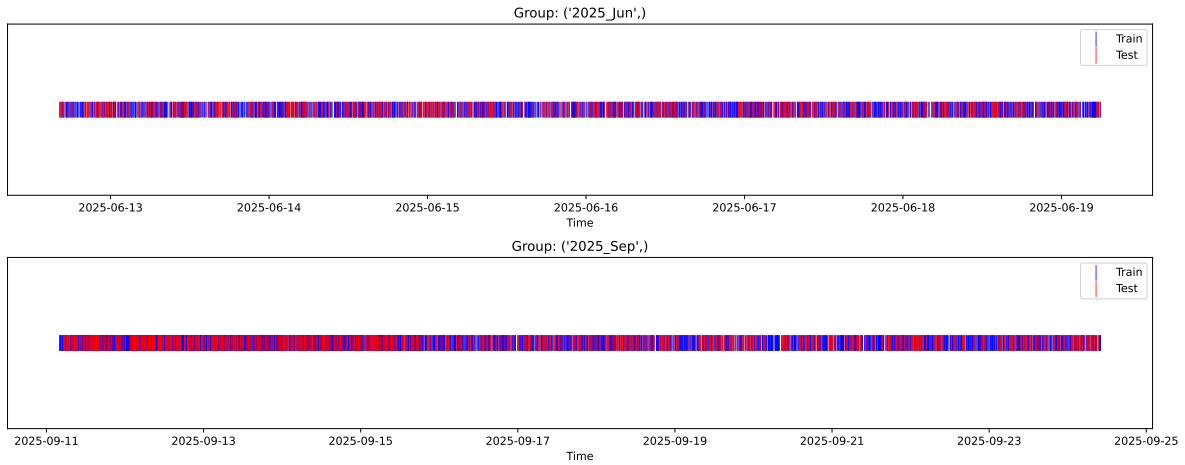
External Split: Train: 5591 samples Test: 2397 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



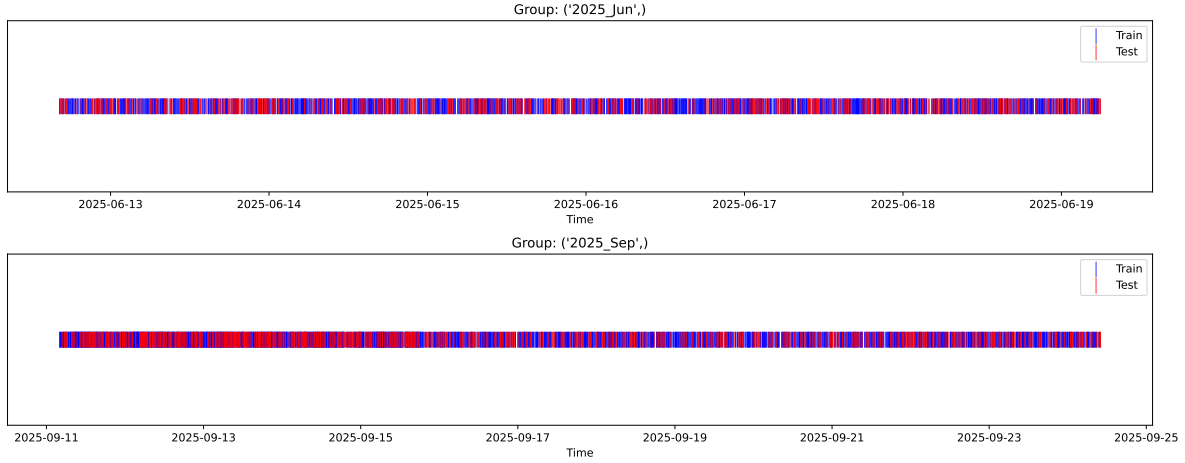
Fold 3/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



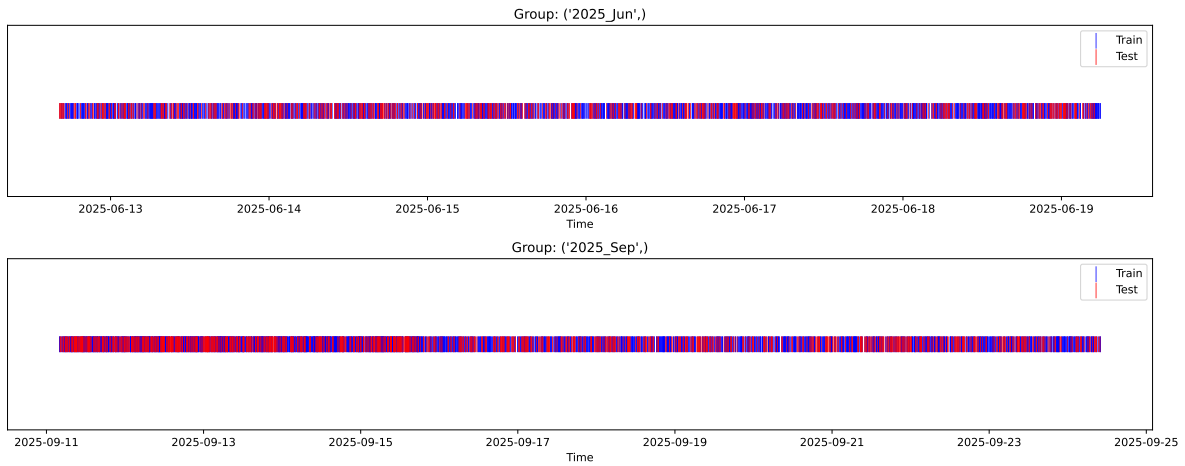
Fold 4/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.350, std=0.793, min=0.959, max=8.682
 y_{test} : mean=2.359, std=0.815, min=0.973, max=13.845

Fold 0 — y_{test} : mean=2.36, std=0.79, y_{train} : mean=2.35, std=0.79

Fold 1 — y_{test} : mean=2.35, std=0.78, y_{train} : mean=2.35, std=0.80

Fold 2 — y_{test} : mean=2.34, std=0.79, y_{train} : mean=2.35, std=0.79

Fold 3 — y_test: mean=2.37, std=0.83, y_train: mean=2.34, std=0.77

Fold 4 — y_test: mean=2.37, std=0.79, y_train: mean=2.34, std=0.79

train scores CV: [0.50702516 0.48874892 0.4956447 0.49357868 0.50363177]

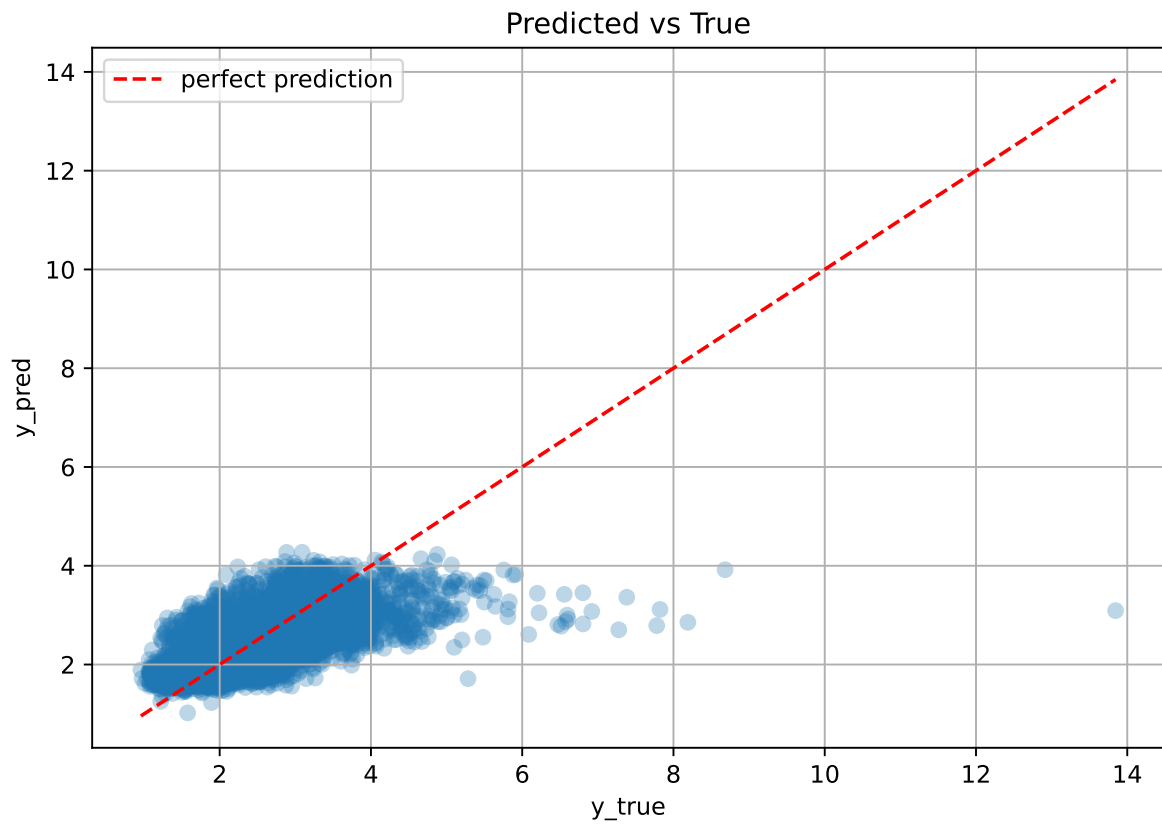
test scores CV: [0.47106436 0.51444549 0.49799754 0.50073954 0.477378]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4977	0.4923	0.0159	0.4664	0.4669	-0.0254	0.9484	1.011

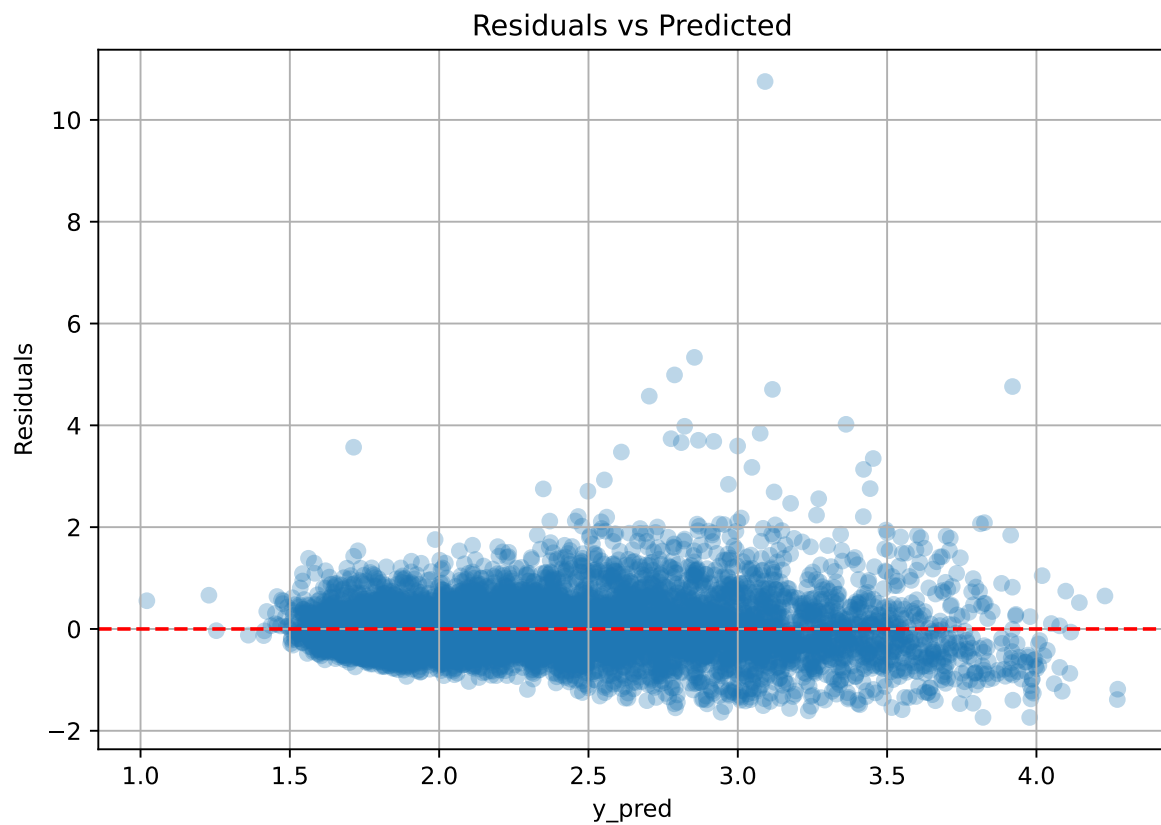
Coefficients:

coef_r_CO2 = -0.0438, coef_r_NH3_st_24h_diff_last = 0.0114, coef_r_NH3_st_24h_median = -0.0611, coef_r_NH3_st_24h_slope = 0.0579, coef_r_NH3_st_24h_std = 0.0757, coef_r_temperature_st_24h_diff_last = 0.1937, coef_r_umidity_st_24h_diff_last = -0.1222, coef_r_umidity_st_24h_max = -0.2879, coef_r_umidity_st_24h_slope = -0.2334, coef_r_umidity_st_24h_std = 0.1374, intercept = 2.3500,

3.4.2 Scatter



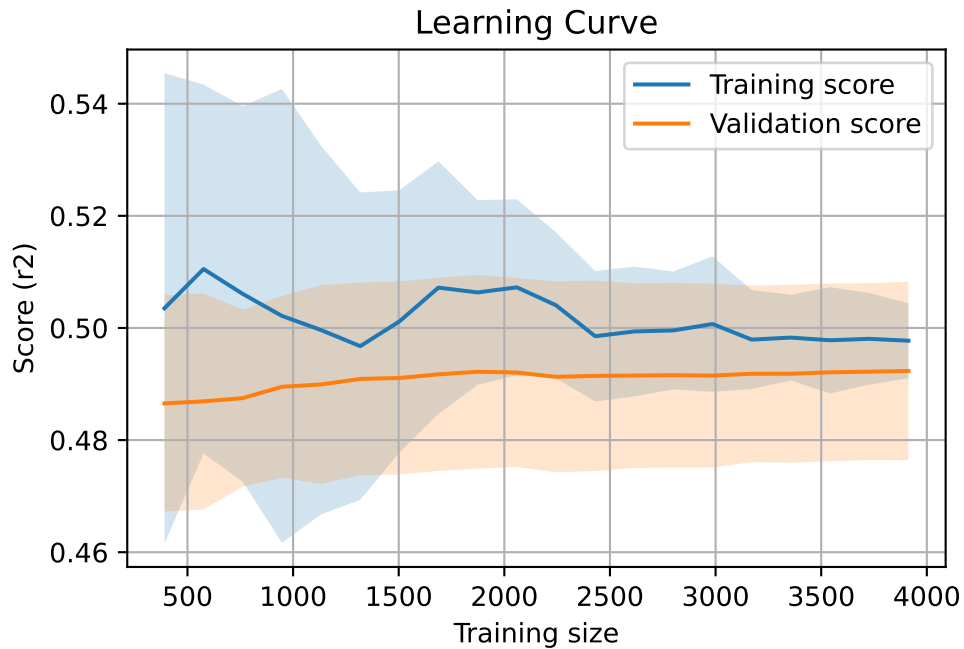
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396

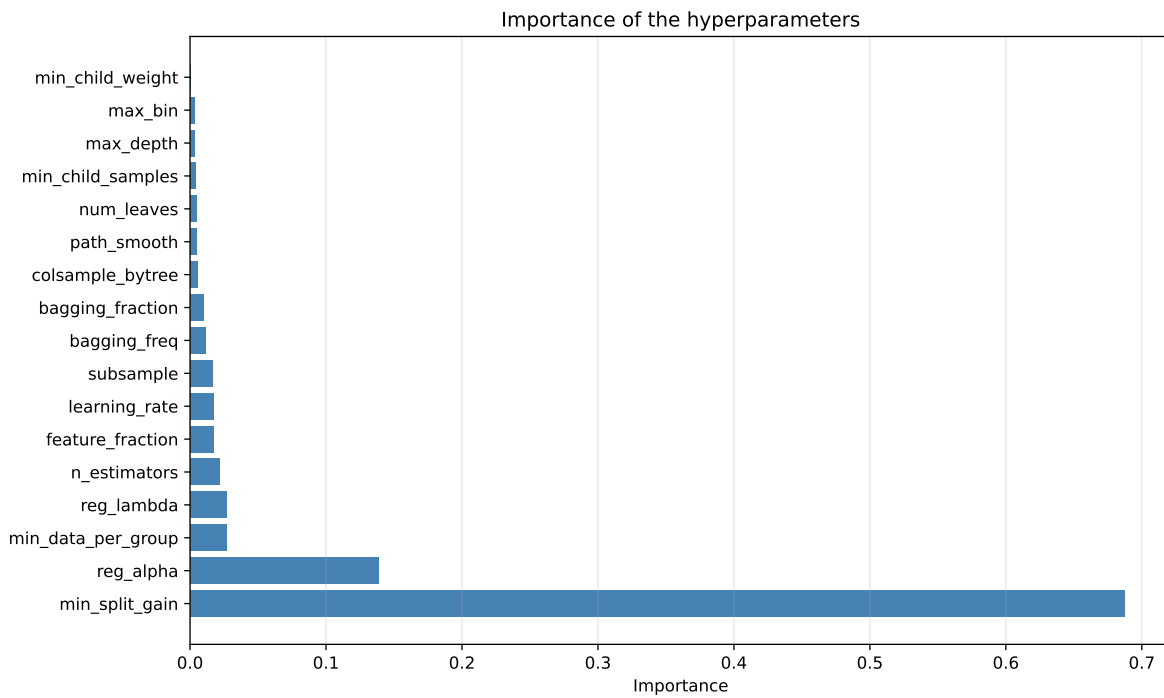
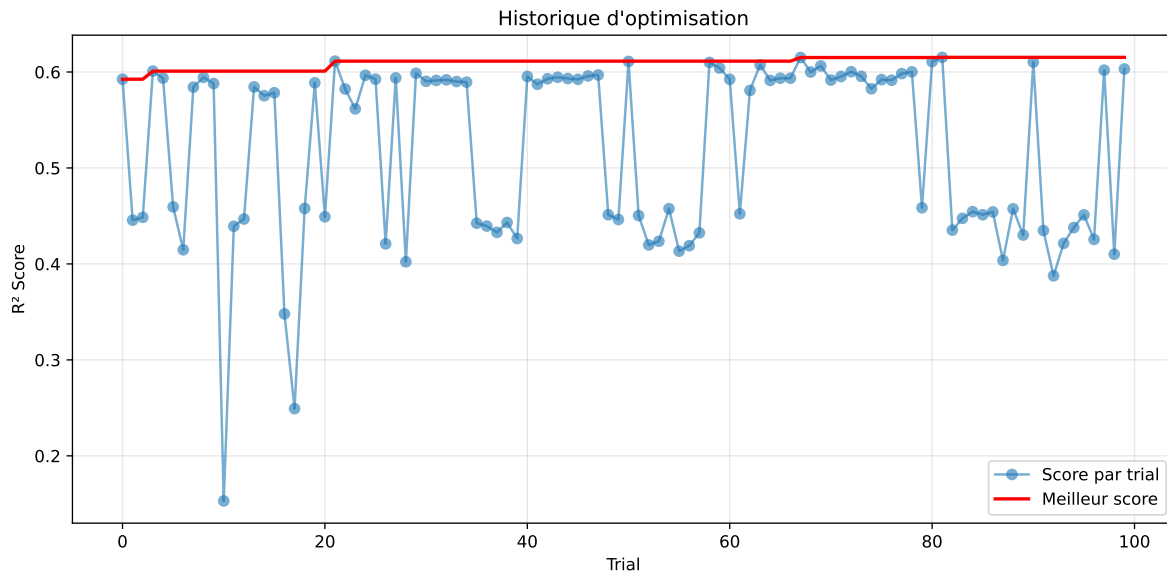


3.5 Model : LGBM

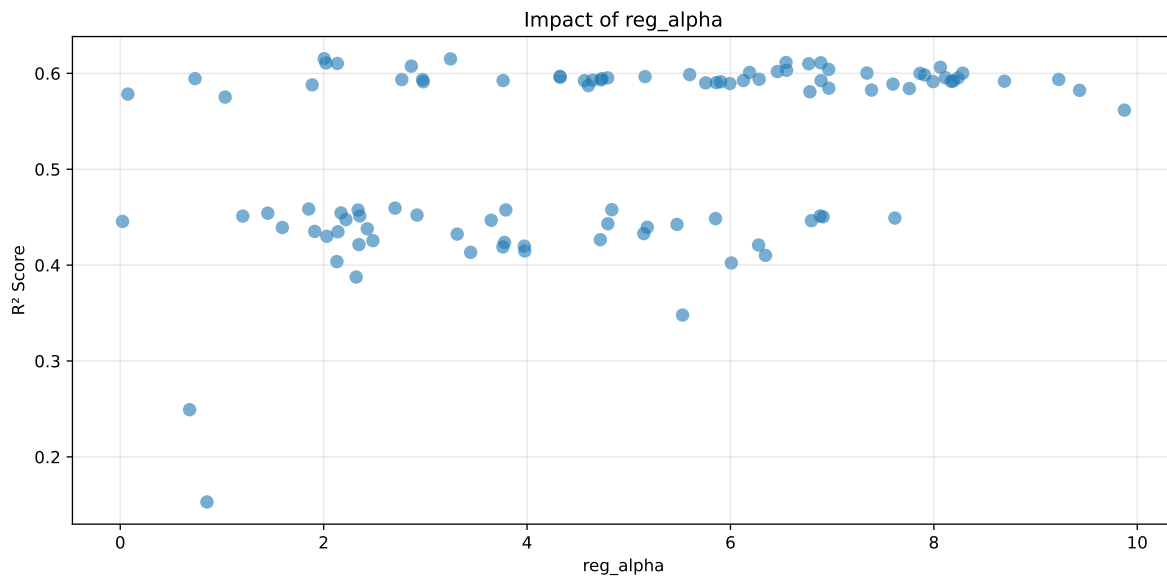
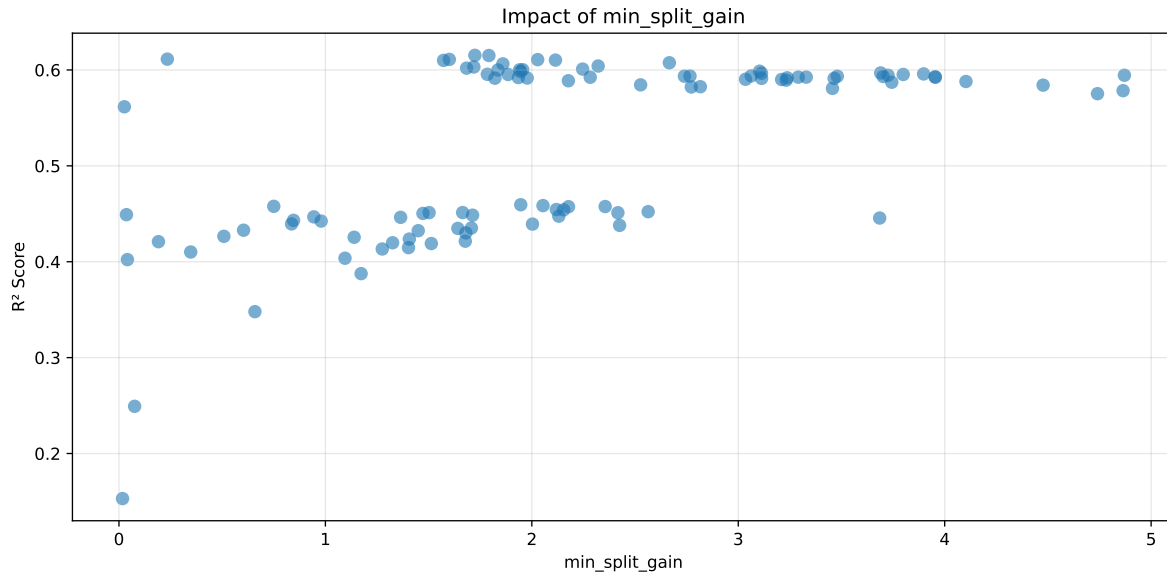
3.5.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.350, std=0.793, min=0.959, max=8.682
y_test: mean=2.359, std=0.815, min=0.973, max=13.845

===== TOP Importance FEATURES =====
feature importance r_umidity_st_24h_max 94 r_CO2 86 r_NH3_st_24h_median 71
r_umidity_st_24h_slope 54 r_temperature_st_24h_diff_last 53 r_NH3_st_24h_std
52 r_umidity_st_24h_diff_last 32 r_NH3_st_24h_slope 31 r_umidity_st_24h_std 30
r_NH3_st_24h_diff_last 22

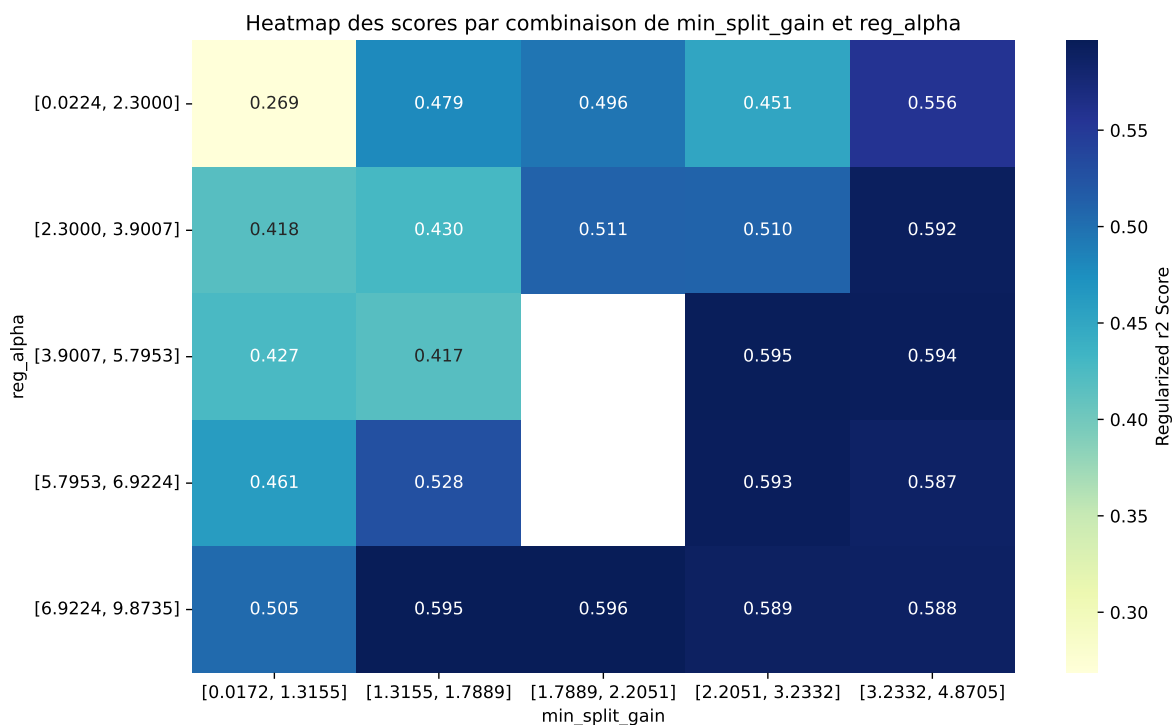


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6457	0.6153	0.603	0.0138	0.6153	0.6193	0.004	1.0065	1.0495
2	0.6458	0.6151	0.6028	0.0152	0.6151	0.6116	-0.0035	0.9944	1.05
3	0.6414	0.6113	0.5981	0.0142	0.6113	0.6053	-0.006	0.9902	1.0493
4	0.6404	0.611	0.5966	0.012	0.611	0.608	-0.003	0.995	1.048
5	0.6407	0.6108	0.5982	0.0116	0.6108	0.6122	0.0014	1.0024	1.049
6	0.64	0.6103	0.5978	0.0101	0.6103	0.6078	-0.0025	0.9959	1.0488
7	0.6403	0.61	0.5964	0.0125	0.61	0.6096	-0.0004	0.9993	1.0497
8	0.6377	0.6075	0.5924	0.0139	0.6075	0.6044	-0.0031	0.9949	1.0497
9	0.6333	0.6063	0.5949	0.0147	0.6063	0.6056	-0.0007	0.9989	1.0444
10	0.629	0.6041	0.5903	0.0142	0.6041	0.605	0.0009	1.0015	1.0411
---	---	---	---	---	---	---	---	---	---
100	0.8008	0.6928	0.6937	0.0124	0.153	0.7078	0.0149	1.0216	1.1558

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.06800131482461252, 'reg__num_leaves': 32, 'reg__subsample': 0.7242027810571251, 'reg__colsample_bytree': 0.7472374958070485, 'reg__min_child_samples': 14, 'reg__reg_alpha': 2.0056579274583806, 'reg__reg_lambda': 7.8458599003944585, 'reg__min_split_gain': 1.7240112801300336, 'reg__path_smooth': 4.666380053536747, 'reg__min_child_weight':

0.0014534316611044837, 'reg__feature_fraction': 0.6937208793192027, 'reg__bagging_fraction': 0.5249831565528417, 'reg__bagging_freq': 2, 'reg__max_bin': 156, 'reg__min_data_per_group': 53}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.0554142989435736, 'reg__num_leaves': 24, 'reg__subsample': 0.7214359302323375, 'reg__colsample_bytree': 0.7021574740196284, 'reg__min_child_samples': 17, 'reg__reg_alpha': 3.2473949865023646, 'reg__reg_lambda': 10.67348591376191, 'reg__min_split_gain': 1.7919580072670946, 'reg__path_smooth': 4.548587144262033, 'reg__min_child_weight': 0.004775222696295226, 'reg__feature_fraction': 0.6822161131667273, 'reg__bagging_fraction': 0.6251902382672202, 'reg__bagging_freq': 4, 'reg__max_bin': 192, 'reg__min_data_per_group': 63}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.01219880977070469, 'reg__num_leaves': 16, 'reg__subsample': 0.6687987141935196, 'reg__colsample_bytree': 0.505654448567254, 'reg__min_child_samples': 28, 'reg__reg_alpha': 6.5465693772336, 'reg__reg_lambda': 26.59829924263578, 'reg__min_split_gain': 0.2342726730094089, 'reg__path_smooth': 3.0017985910445635, 'reg__min_child_weight': 0.0010625657709340962, 'reg__feature_fraction': 0.5086480508372767, 'reg__bagging_fraction': 0.5302093347016616, 'reg__bagging_freq': 7, 'reg__max_bin': 229, 'reg__min_data_per_group': 74}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.032437657151281225, 'reg__num_leaves': 29, 'reg__subsample': 0.723073529656304, 'reg__colsample_bytree': 0.6912035911928941, 'reg__min_child_samples': 24, 'reg__reg_alpha': 6.888587227608937, 'reg__reg_lambda': 13.668559820634568, 'reg__min_split_gain': 1.6002676984137039, 'reg__path_smooth': 3.76596096340813, 'reg__min_child_weight': 0.0027628007318099166, 'reg__feature_fraction': 0.6294499217818001, 'reg__bagging_fraction': 0.7021138144067854, 'reg__bagging_freq': 5, 'reg__max_bin': 206, 'reg__min_data_per_group': 67}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.05109118607483221, 'reg__num_leaves': 33, 'reg__subsample': 0.5813148664205282, 'reg__colsample_bytree': 0.5255540782229778, 'reg__min_child_samples': 10, 'reg__reg_alpha': 2.022643152093973, 'reg__reg_lambda': 7.7014200197927725, 'reg__min_split_gain': 2.0282074653527564, 'reg__path_smooth': 4.777474118272762, 'reg__min_child_weight': 0.0015865091185397096, 'reg__feature_fraction': 0.698948602642365, 'reg__bagging_fraction': 0.5245441361104628, 'reg__bagging_freq': 2, 'reg__max_bin': 153, 'reg__min_data_per_group': 57}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.06549289140035174, 'reg__num_leaves': 38, 'reg__subsample': 0.5685112553386212, 'reg__colsample_bytree': 0.5625442077222296, 'reg__min_child_samples': 10, 'reg__reg_alpha': 2.1342376157263265, 'reg__reg_lambda': 8.712755032225765, 'reg__min_split_gain': 2.114504906019917, 'reg__path_smooth': 4.131543855974664, 'reg__min_child_weight':

0.0014650462059406797, 'reg__feature_fraction': 0.7027924720476412, 'reg__bagging_fraction': 0.5676057811218189, 'reg__bagging_freq': 3, 'reg__max_bin': 138, 'reg__min_data_per_group': 70}

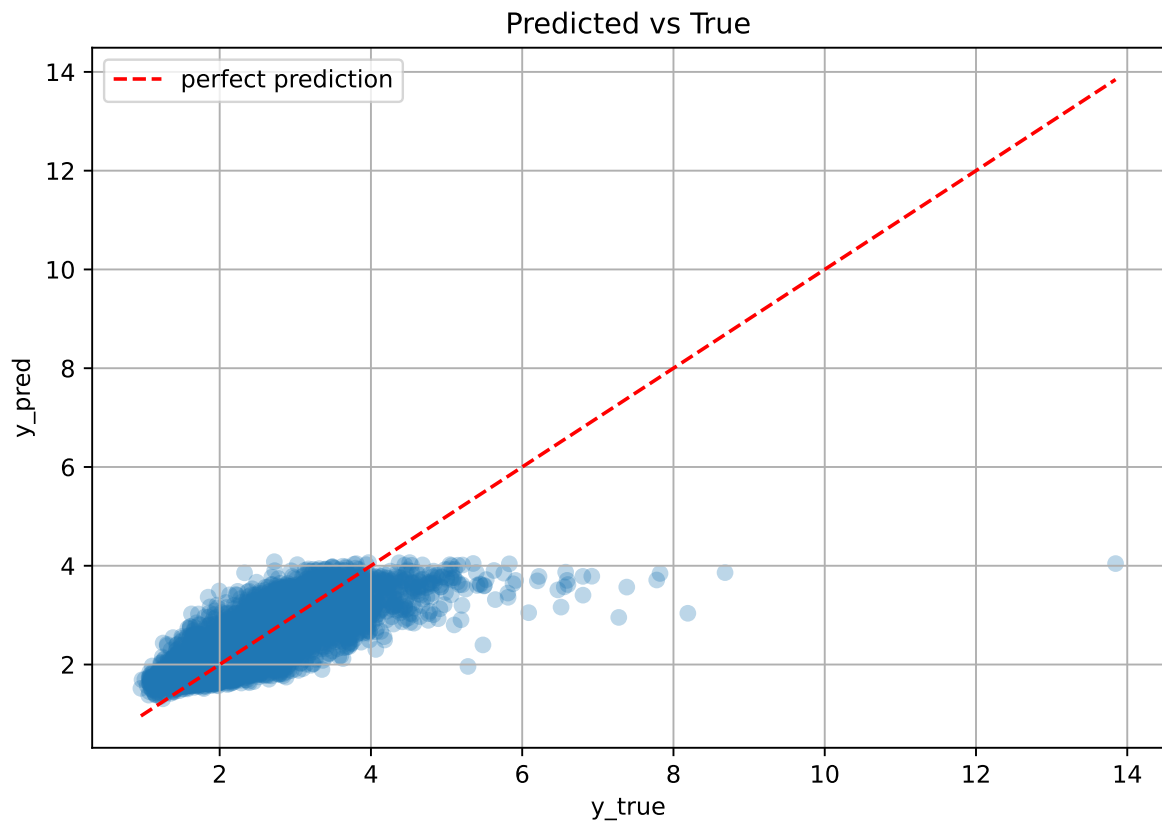
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.034884597058917166, 'reg__num_leaves': 29, 'reg__subsample': 0.6796674940132491, 'reg__colsample_bytree': 0.691428699416755, 'reg__min_child_samples': 24, 'reg__reg_alpha': 6.769987158023062, 'reg__reg_lambda': 15.788489253946592, 'reg__min_split_gain': 1.572290356836636, 'reg__path_smooth': 4.217129840655168, 'reg__min_child_weight': 0.0023602815716876934, 'reg__feature_fraction': 0.5248336653244438, 'reg__bagging_fraction': 0.7069486876341013, 'reg__bagging_freq': 4, 'reg__max_bin': 205, 'reg__min_data_per_group': 61}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.041197979009839715, 'reg__num_leaves': 23, 'reg__subsample': 0.6178707047766754, 'reg__colsample_bytree': 0.5314371213957305, 'reg__min_child_samples': 17, 'reg__reg_alpha': 2.8623099402743946, 'reg__reg_lambda': 10.536899645401128, 'reg__min_split_gain': 2.666187579836573, 'reg__path_smooth': 4.335997468362648, 'reg__min_child_weight': 0.0010261370449913053, 'reg__feature_fraction': 0.6464652332243065, 'reg__bagging_fraction': 0.8209454041530215, 'reg__bagging_freq': 4, 'reg__max_bin': 183, 'reg__min_data_per_group': 63}

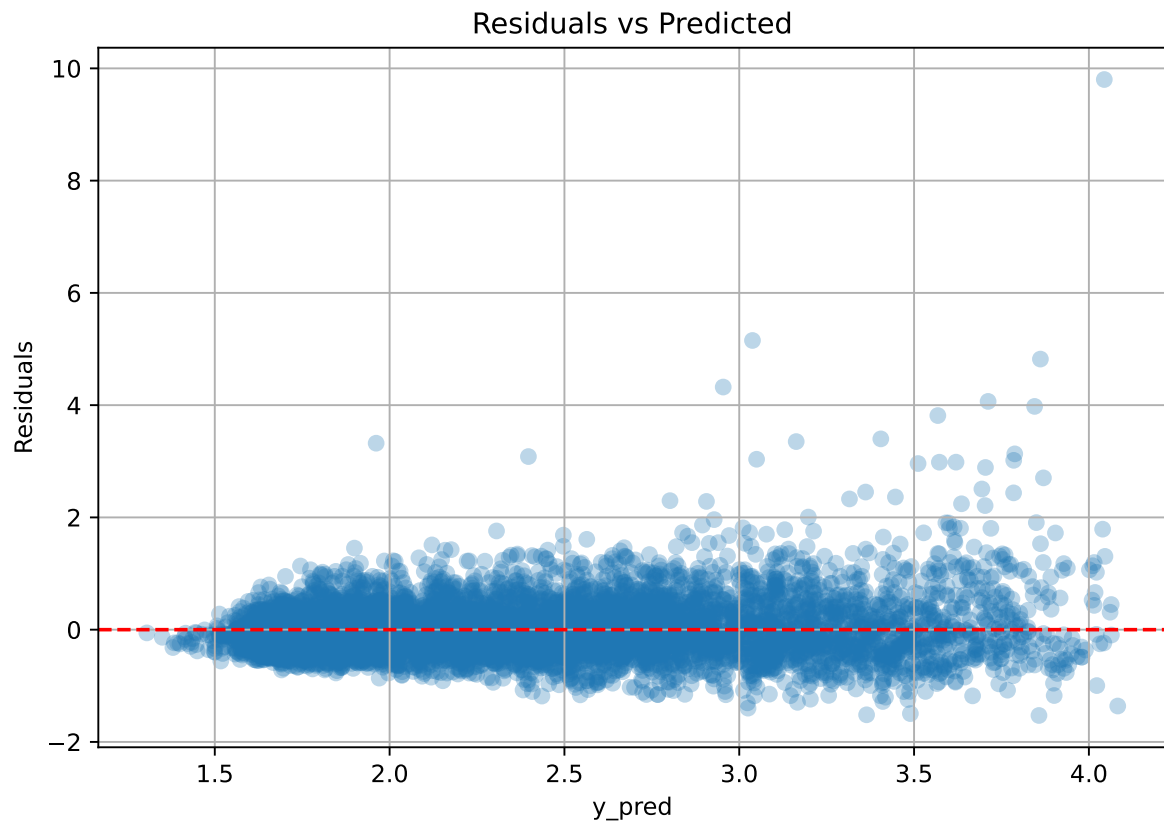
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.05949416252393842, 'reg__num_leaves': 23, 'reg__subsample': 0.611547690510713, 'reg__colsample_bytree': 0.6170420471066679, 'reg__min_child_samples': 18, 'reg__reg_alpha': 8.06410817455823, 'reg__reg_lambda': 10.291827245453309, 'reg__min_split_gain': 1.8596709666090645, 'reg__path_smooth': 4.950782117709249, 'reg__min_child_weight': 0.004492816750381966, 'reg__feature_fraction': 0.5619229847441685, 'reg__bagging_fraction': 0.5101750958192088, 'reg__bagging_freq': 1, 'reg__max_bin': 188, 'reg__min_data_per_group': 57}

Rank 10: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.04069227221857748, 'reg__num_leaves': 27, 'reg__subsample': 0.6815708381816261, 'reg__colsample_bytree': 0.6910458657895895, 'reg__min_child_samples': 17, 'reg__reg_alpha': 6.9660181005706825, 'reg__reg_lambda': 34.69811359975343, 'reg__min_split_gain': 2.3213668349889387, 'reg__path_smooth': 4.547773880912541, 'reg__min_child_weight': 0.002093551246184468, 'reg__feature_fraction': 0.5606266843137331, 'reg__bagging_fraction': 0.615288402558053, 'reg__bagging_freq': 1, 'reg__max_bin': 206, 'reg__min_data_per_group': 64}

3.5.2 Scatter



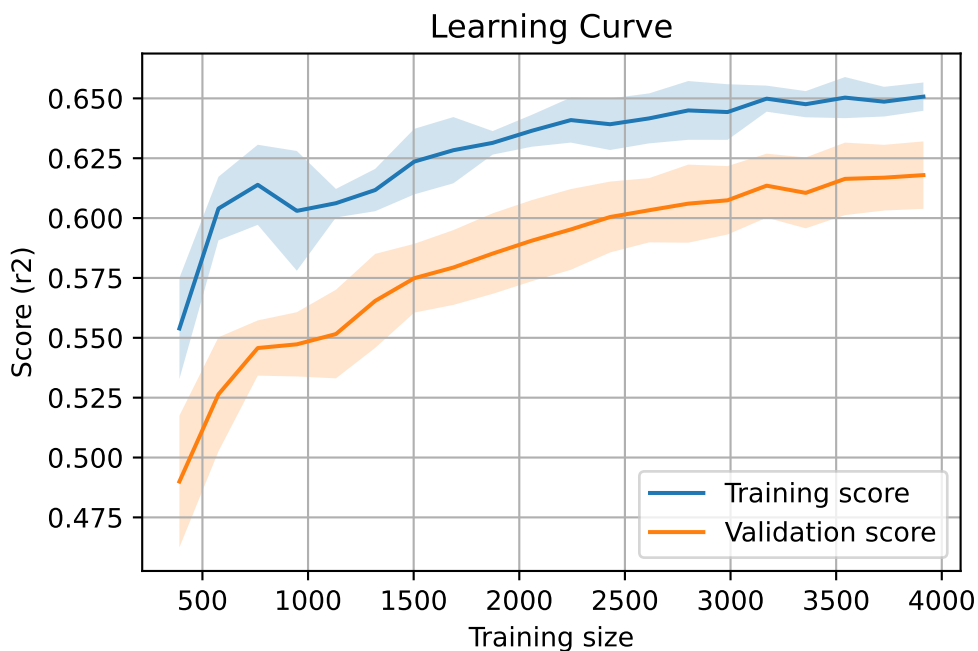
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396

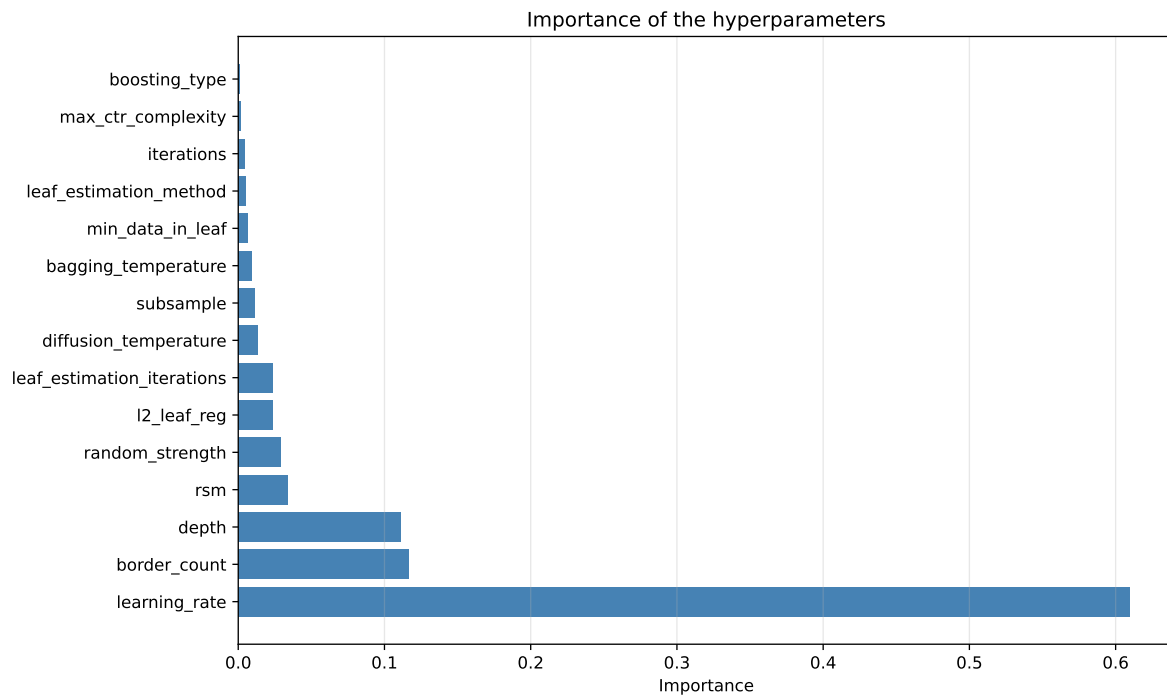
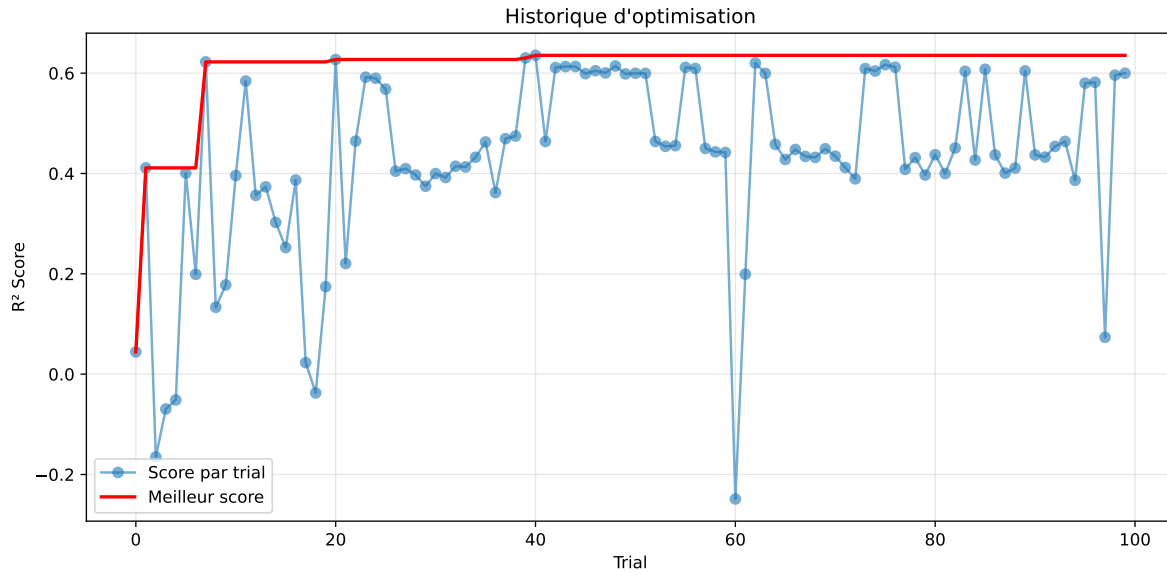


3.6 Model : CatBoost

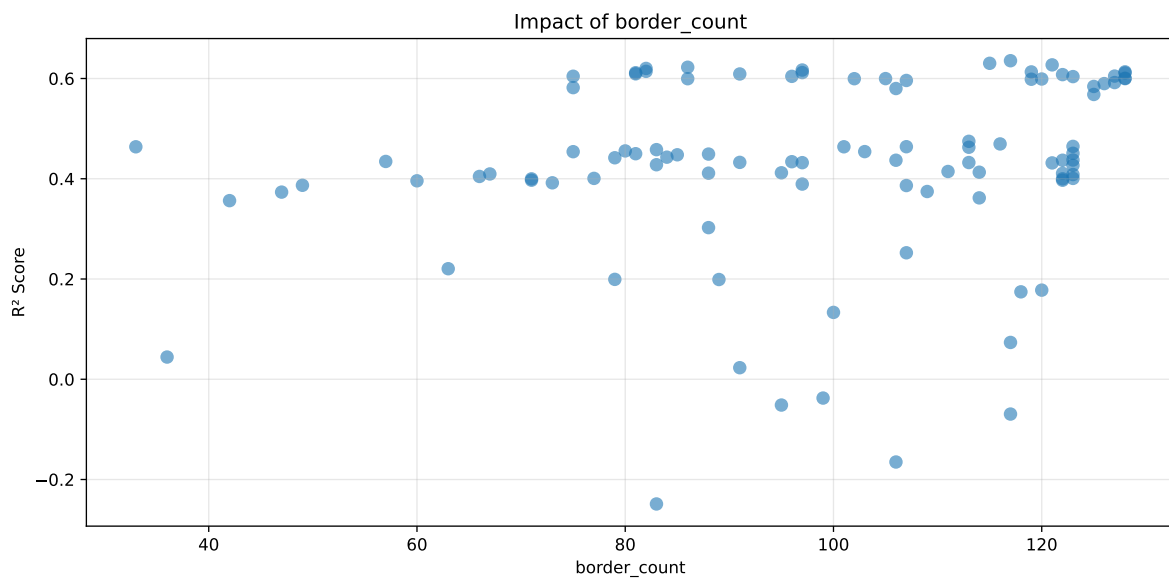
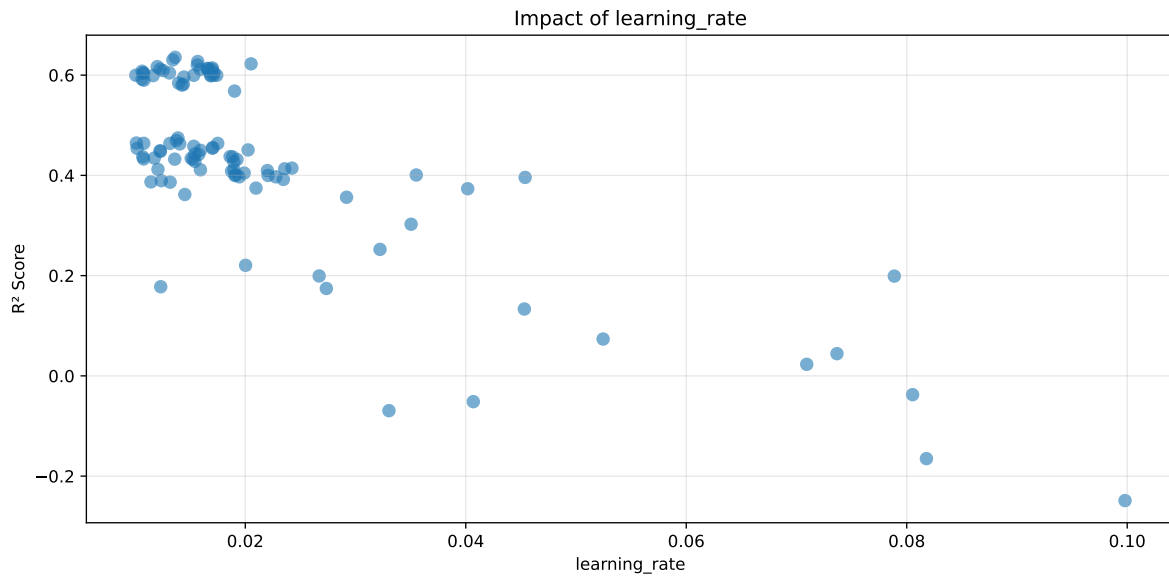
3.6.1 Gridsearch

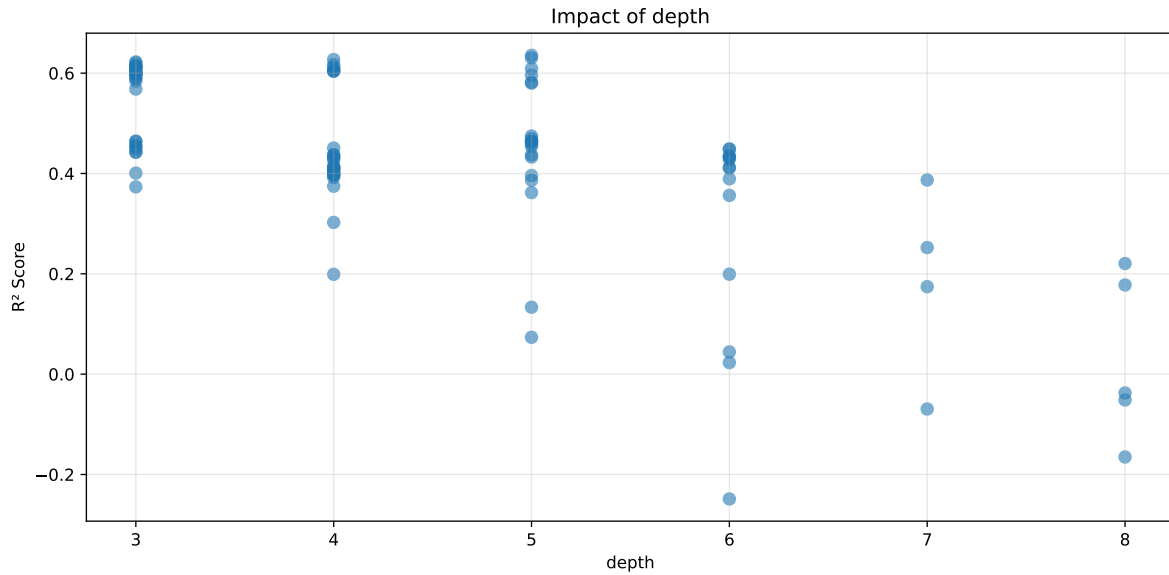
Distribution y_train vs y_test: y_train: mean=2.350, std=0.793, min=0.959, max=8.682
y_test: mean=2.359, std=0.815, min=0.973, max=13.845

===== TOP Importance FEATURES =====
feature importance r_umidity_st_24h_max 22.338813 r_umidity_st_24h_slope 21.154823
r_temperature_st_24h_diff_last 17.267161 r_CO2 9.008393 r_NH3_st_24h_std
6.953228 r_NH3_st_24h_median 6.308358 r_umidity_st_24h_std 6.006686 r_umid-
ity_st_24h_diff_last 4.649685 r_NH3_st_24h_slope 4.486630 r_NH3_st_24h_diff_last
1.826222



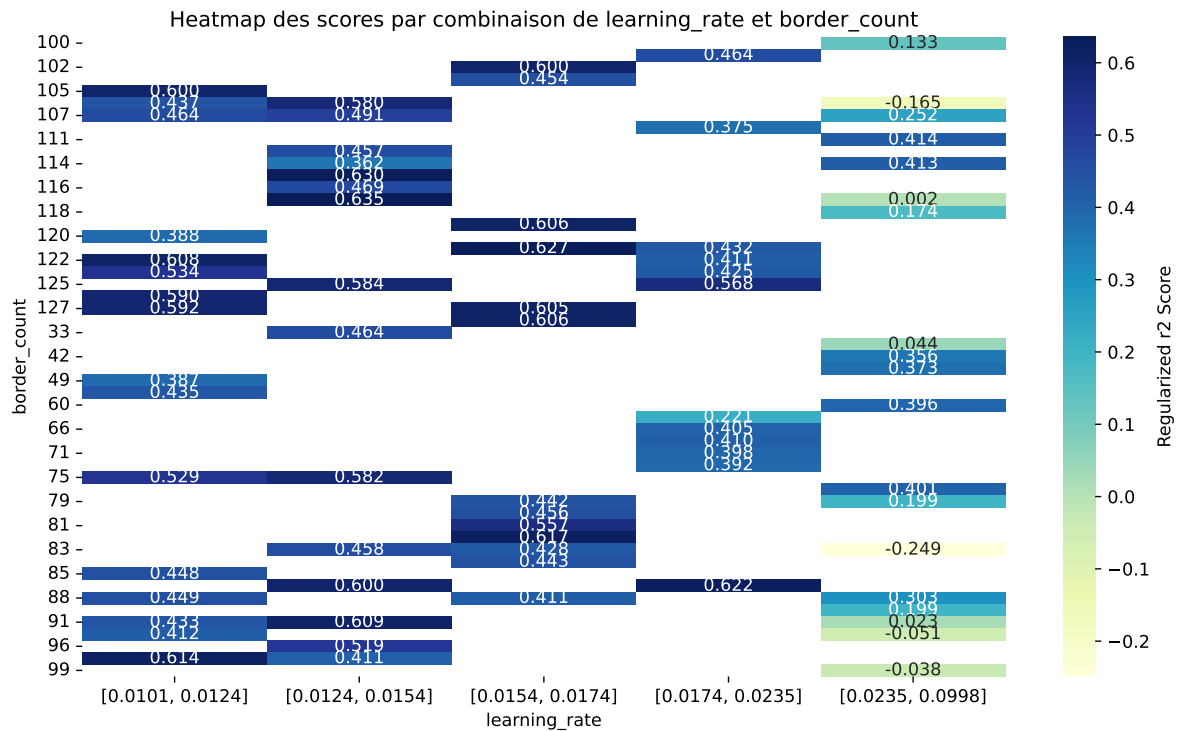
Paramètres avec >5% d'importance : ['learning_rate', 'border_count', 'depth']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.667	0.6355	0.6224	0.0159	0.6355	0.6305	-0.005	0.9922	1.0496
2	0.6613	0.6303	0.6191	0.0157	0.6303	0.6276	-0.0027	0.9956	1.0491
3	0.655	0.6272	0.6158	0.0156	0.6272	0.62	-0.0072	0.9885	1.0444
4	0.65	0.6225	0.6123	0.0169	0.6225	0.6165	-0.006	0.9904	1.0442
5	0.6508	0.6201	0.6093	0.0152	0.6201	0.6105	-0.0096	0.9845	1.0494
6	0.6426	0.617	0.6039	0.0156	0.617	0.6071	-0.0099	0.984	1.0415
7	0.6444	0.6143	0.5995	0.0159	0.6143	0.6045	-0.0098	0.984	1.049
8	0.6391	0.6133	0.5999	0.0162	0.6133	0.6004	-0.0129	0.979	1.0421
9	0.6406	0.6131	0.6011	0.0163	0.6131	0.6021	-0.011	0.982	1.0448
10	0.6366	0.6118	0.5987	0.0159	0.6118	0.6024	-0.0094	0.9846	1.0406
---	---	---	---	---	---	---	---	---	---
100	0.9359	0.7385	0.7281	0.015	-0.2488	0.7466	0.0081	1.011	1.2674

Hyperparameters:

Rank 1: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.013634948964345642, 'reg__l2_leaf_reg': 35.483283476675176, 'reg__bagging_temperature': 0.538796427208591, 'reg__random_strength': 2.942890293268354, 'reg__subsample': 0.7990491237135213, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5640137096399019, 'reg__border_count': 117, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2864.4014772612327}

Rank 2: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.013428594077152954, 'reg__l2_leaf_reg': 46.271289537996324, 'reg__bagging_temperature': 0.7877063113529916, 'reg__random_strength': 2.8707757684351205, 'reg__subsample': 0.8929354830067053, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5358397939826254, 'reg__border_count': 115, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 8272.282386865081}

Rank 3: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.01568658480525174, 'reg__l2_leaf_reg': 32.90193762659129, 'reg__bagging_temperature': 4.77213949260107, 'reg__random_strength': 4.2810553036454175, 'reg__subsample': 0.5476354755956795, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5759057891885201, 'reg__border_count': 121, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5239.500004684174}

Rank 4: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.020523522275245742, 'reg__l2_leaf_reg': 24.439806401619876, 'reg__bagging_temperature': 0.6773429929949271, 'reg__random_strength': 0.9185172607160469, 'reg__subsample': 0.7458431086970064, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 1, 'reg__rsm':

0.5332596813751354, 'reg__border_count': 86, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8612.707704154896}

Rank 5: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.015650223777747378, 'reg__l2_leaf_reg': 13.3390009830626, 'reg__bagging_temperature': 0.9442424029112496, 'reg__random_strength': 0.5808490126016344, 'reg__subsample': 0.8760198492812366, 'reg__min_data_in_leaf': 19, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.5951769010539049, 'reg__border_count': 82, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6904.879280527075}

Rank 6: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.012016616820067747, 'reg__l2_leaf_reg': 26.814201381964192, 'reg__bagging_temperature': 0.7530832785232942, 'reg__random_strength': 4.982058374383214, 'reg__subsample': 0.8391576069535501, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.5608172845034561, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 9981.64897753375}

Rank 7: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.01701710493001639, 'reg__l2_leaf_reg': 8.104863402648327, 'reg__bagging_temperature': 0.7072236934715466, 'reg__random_strength': 0.9069779734176233, 'reg__subsample': 0.7057089027019218, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5250563182880236, 'reg__border_count': 82, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5290.407367680454}

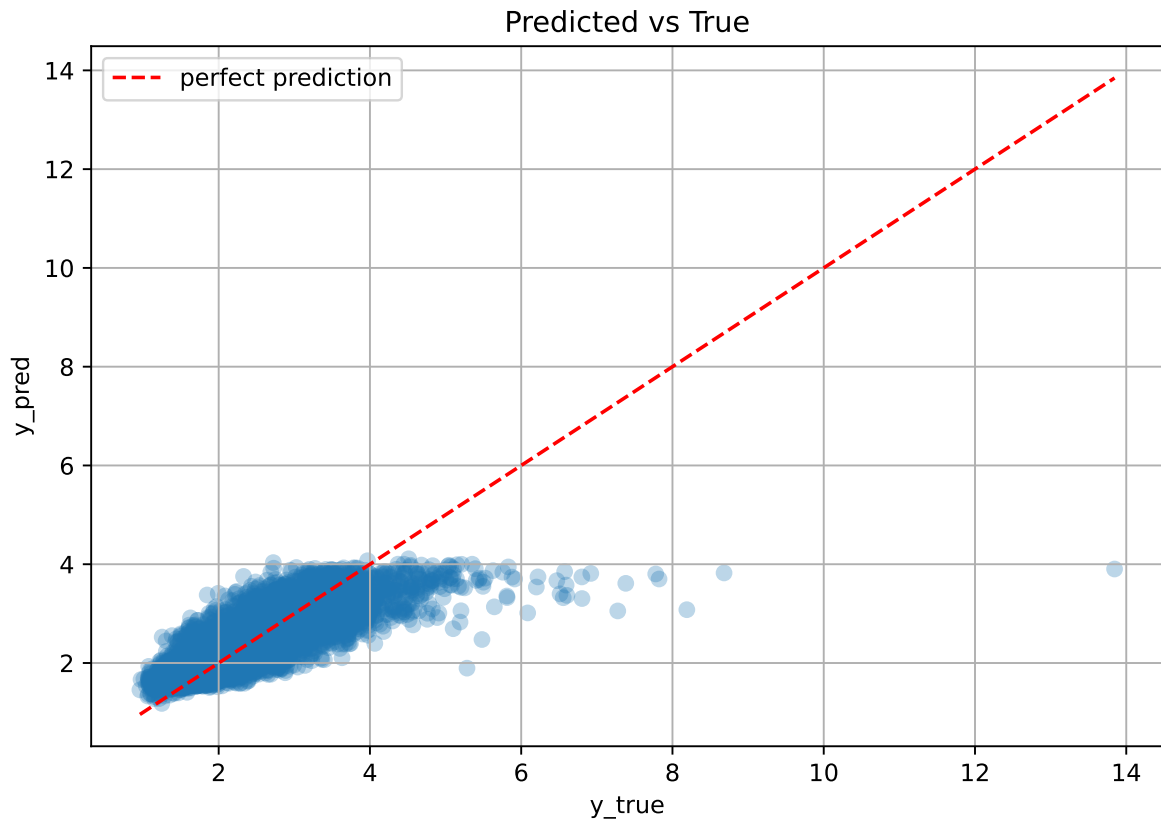
Rank 8: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.016622425083448942, 'reg__l2_leaf_reg': 6.731249386010322, 'reg__bagging_temperature': 0.5466780467416739, 'reg__random_strength': 0.8328629214200665, 'reg__subsample': 0.7201155834052142, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5374090217338321, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 7780.930407593918}

Rank 9: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.016587096764084635, 'reg__l2_leaf_reg': 5.39332474725564, 'reg__bagging_temperature': 0.42674407938952785, 'reg__random_strength': 0.7365092484859532, 'reg__subsample': 0.8837075460904289, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.529259360058894, 'reg__border_count': 119, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 7524.7277474311195}

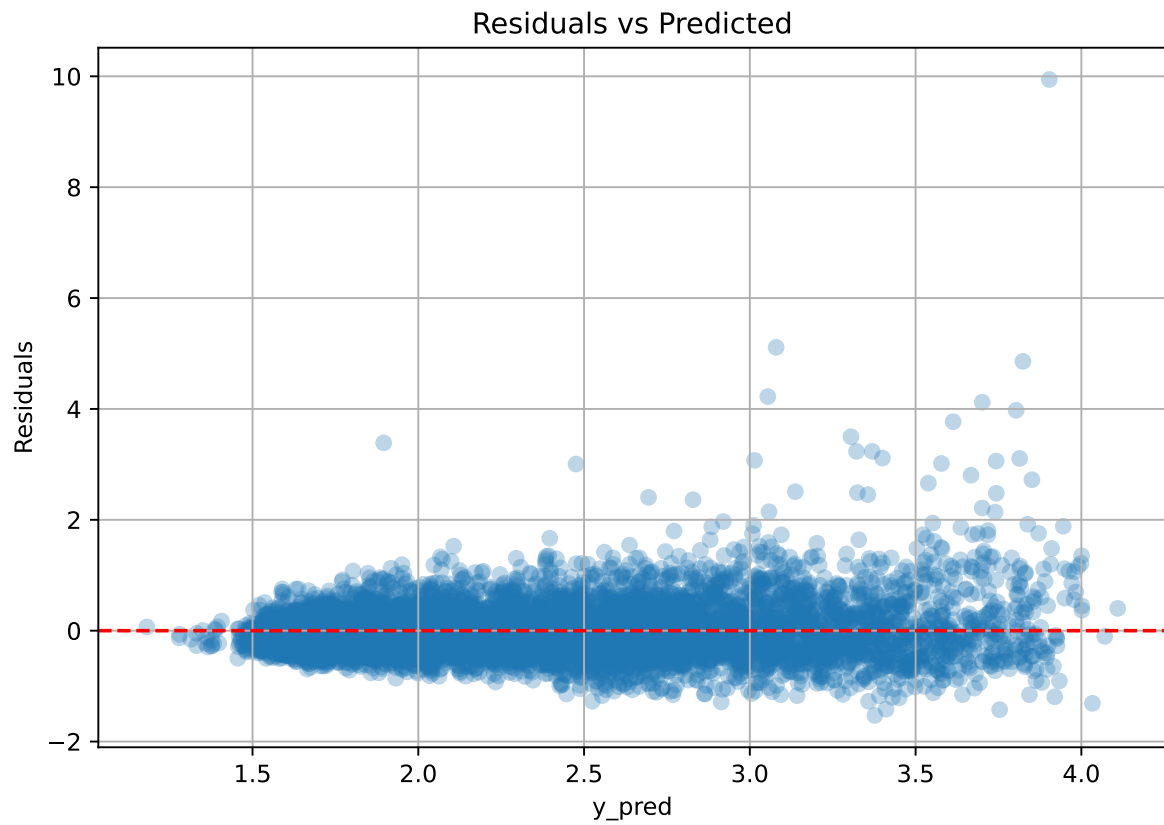
Rank 10: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.012275474173084718, 'reg__l2_leaf_reg': 33.51637826059875, 'reg__bagging_temperature': 0.7579627989157701,

'reg__random_strength': 2.6409406978348913, 'reg__subsample': 0.8314520418534154,
'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 3, 'reg__rsm':
0.5626524154313434, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Newton',
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_tempera-
ture': 85.14517512263592}

3.6.2 Scatter



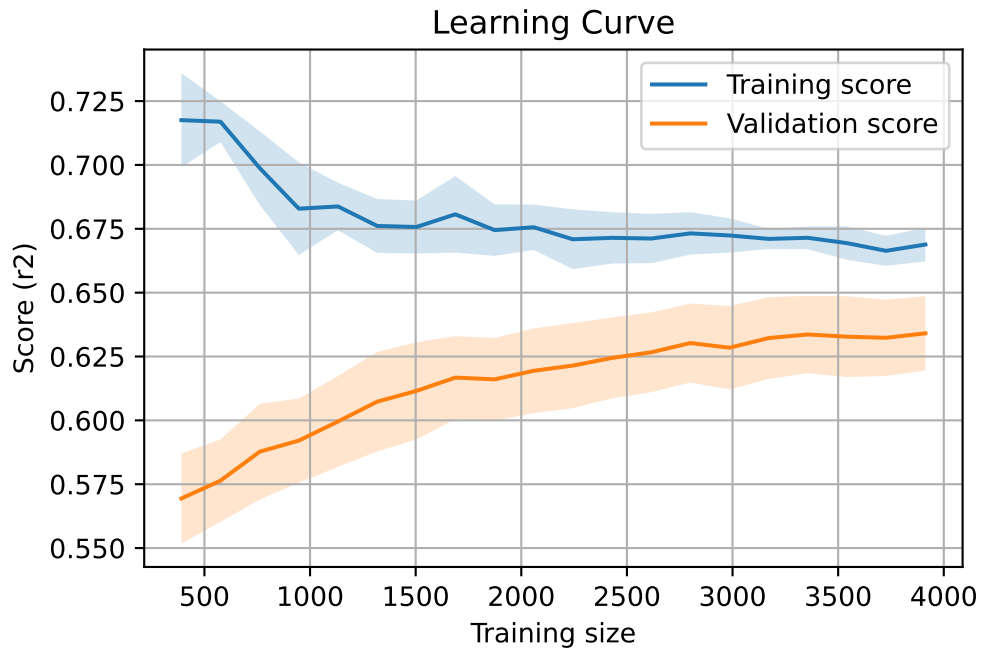
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.498	0.492	0.016	0.466	0.467	-0.025	0.948	1.011	nan
LGBM	0.646	0.615	0.014	0.603	0.619	0.004	1.006	1.05	0.6153
CatBoost	0.667	0.636	0.016	0.622	0.63	-0.005	0.992	1.05	0.6355

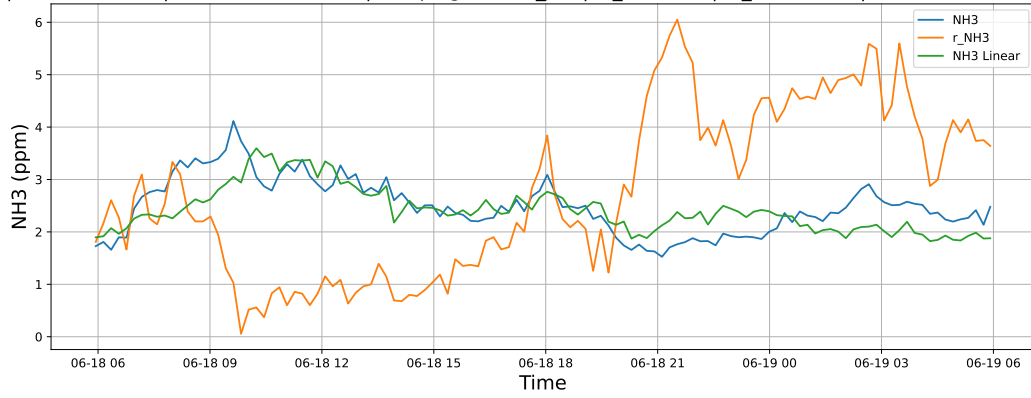
4 Plot

4.1 Standalone plots

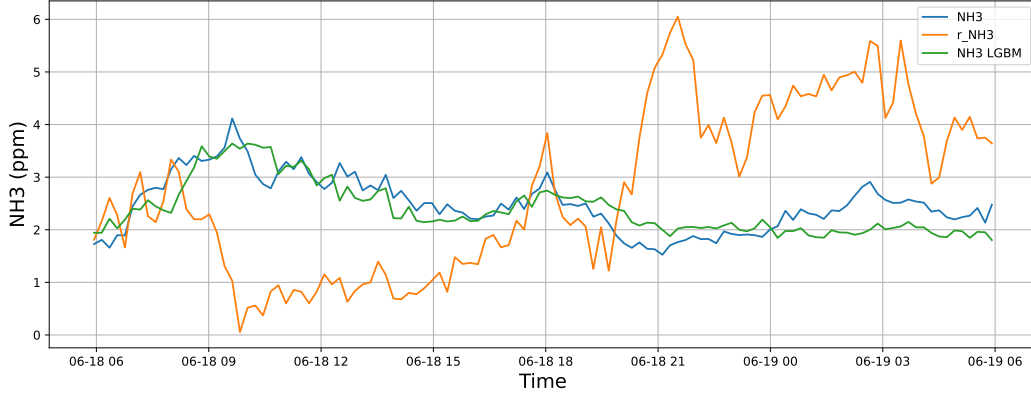
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

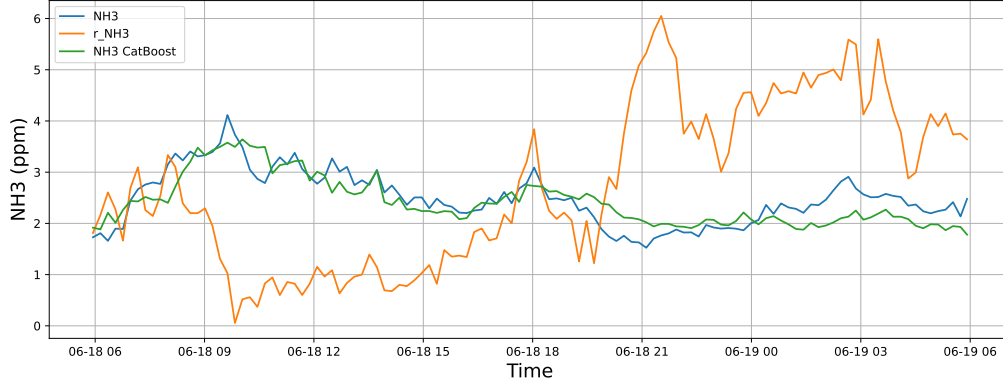
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

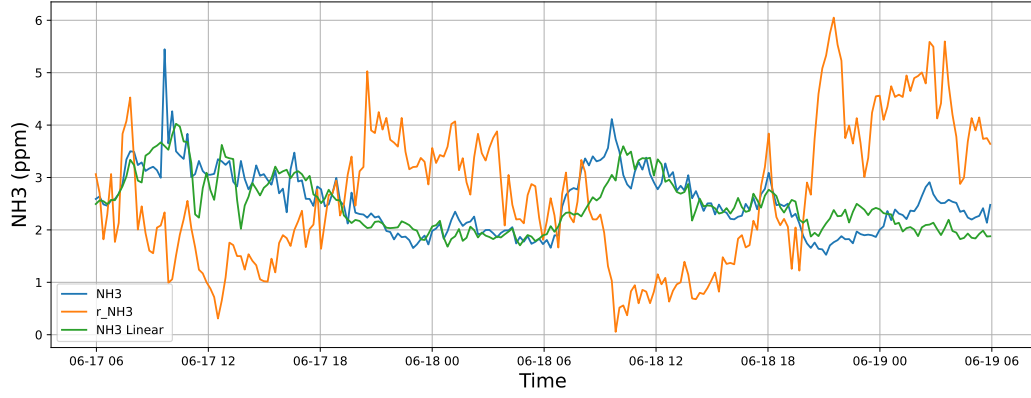


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

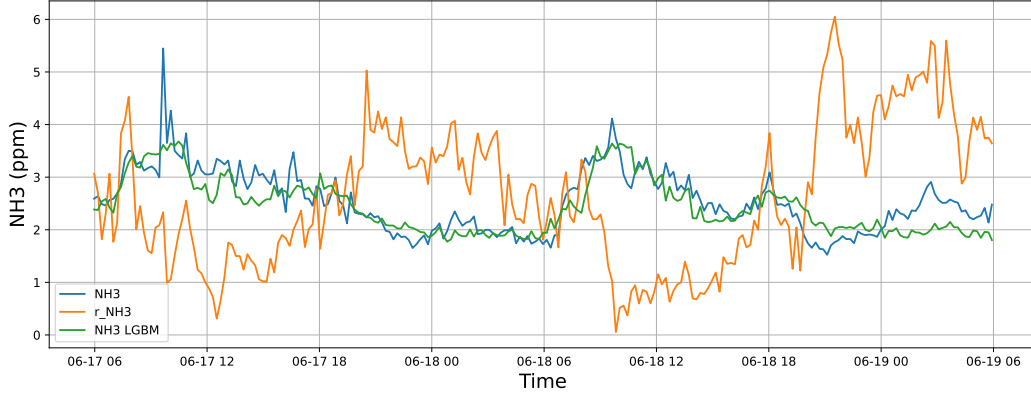


4.1.1.2 Time window : 48

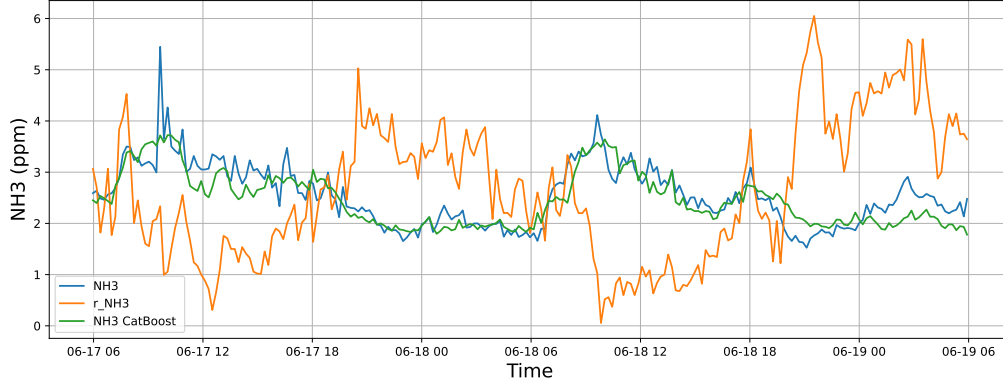
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

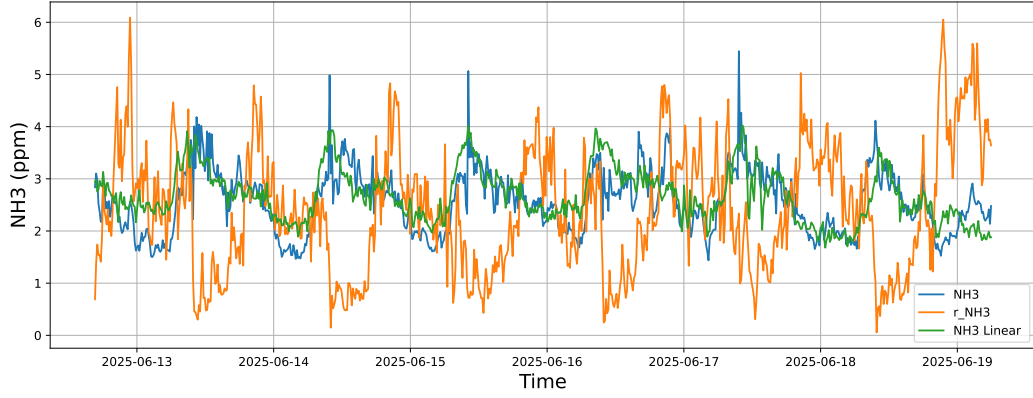


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

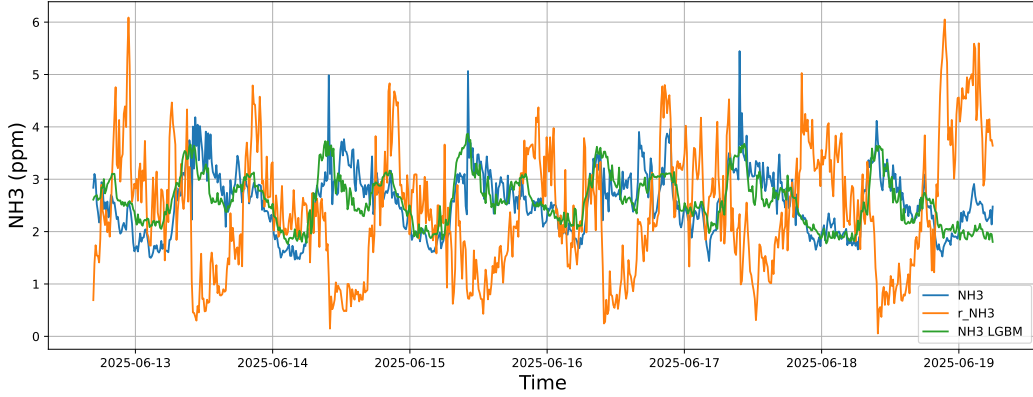


4.1.1.3 Time window : ALL

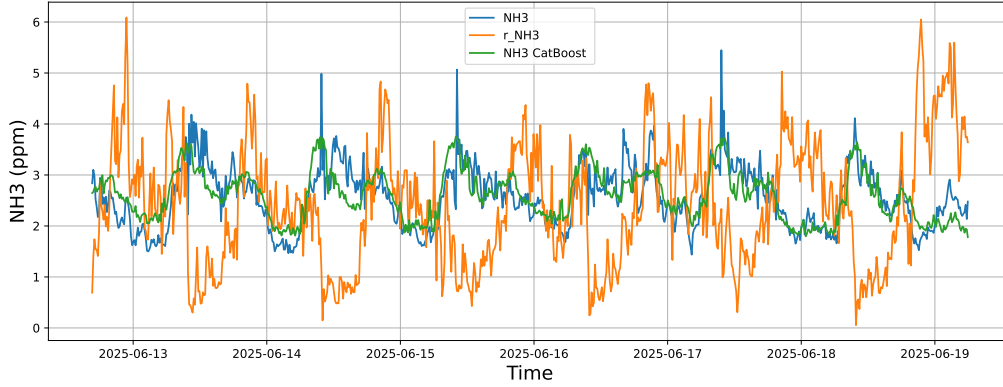
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



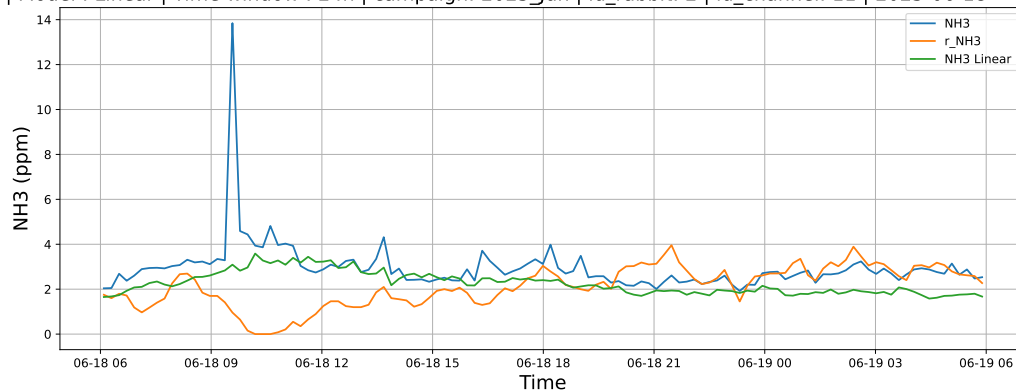
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



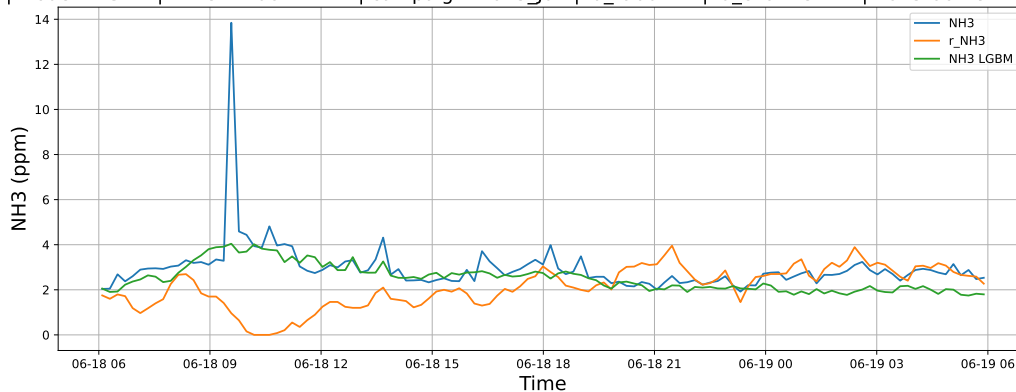
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

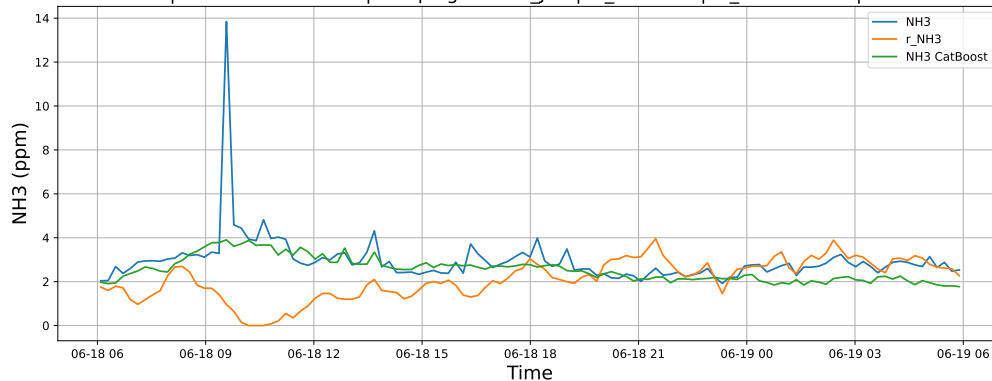
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

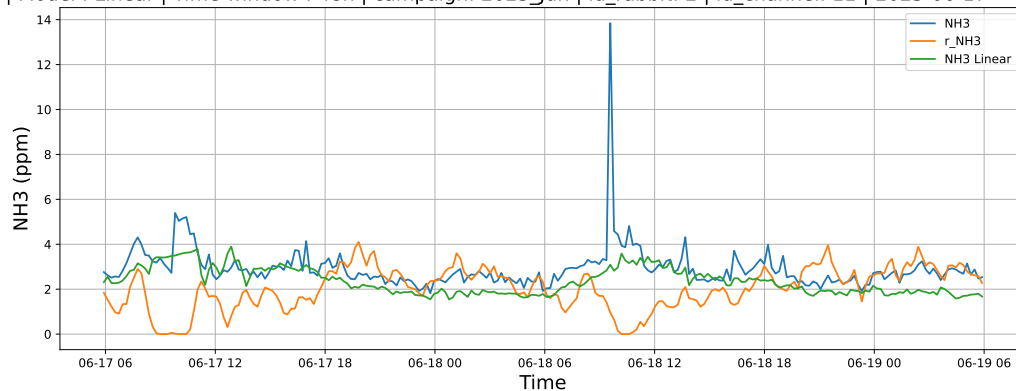


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

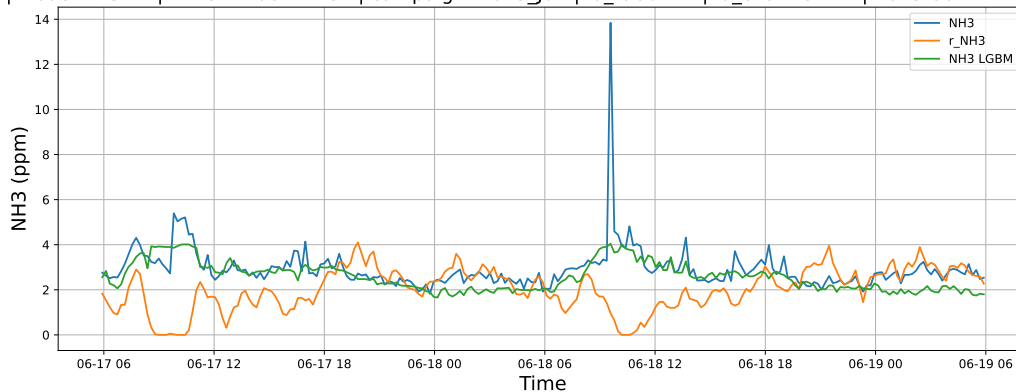


4.1.2.2 Time window : 48

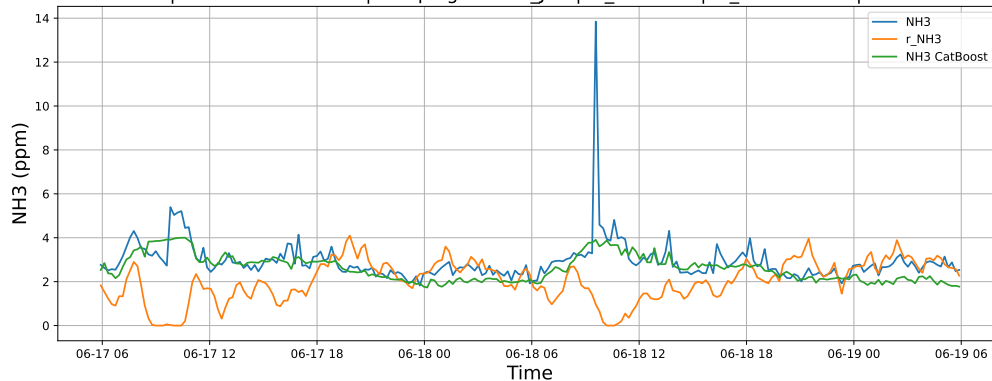
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

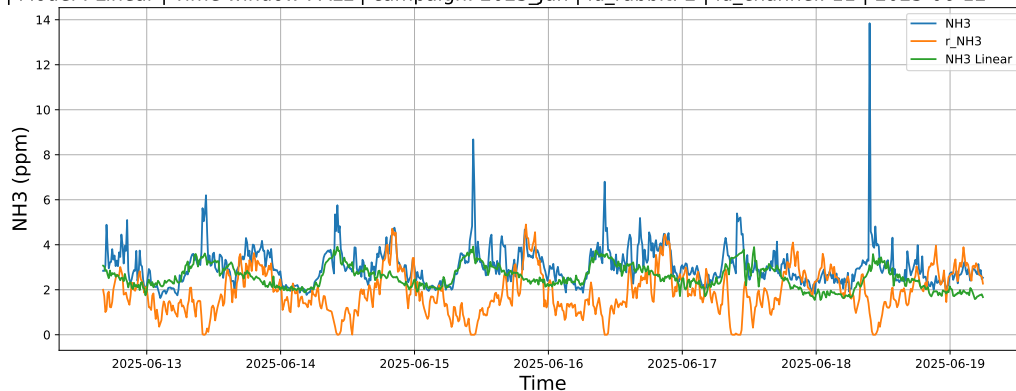


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

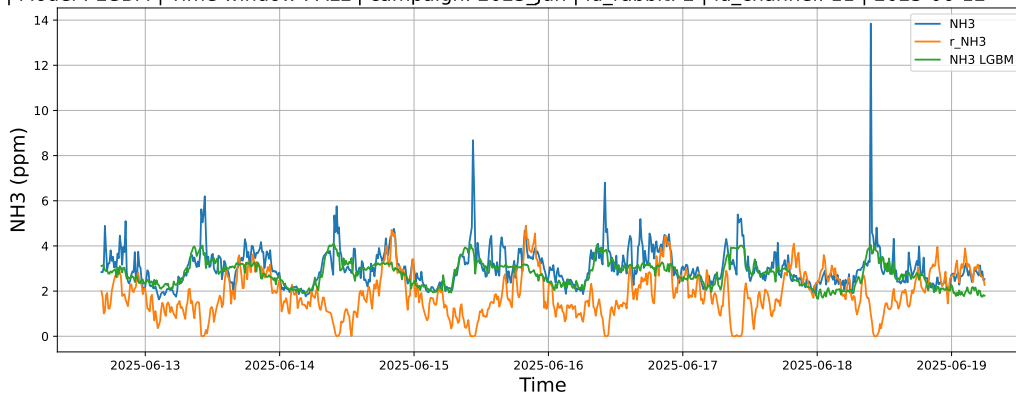


4.1.2.3 Time window : ALL

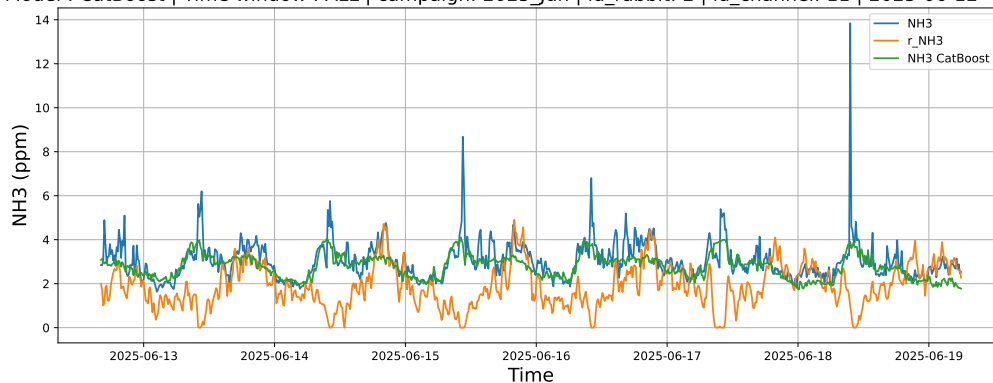
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



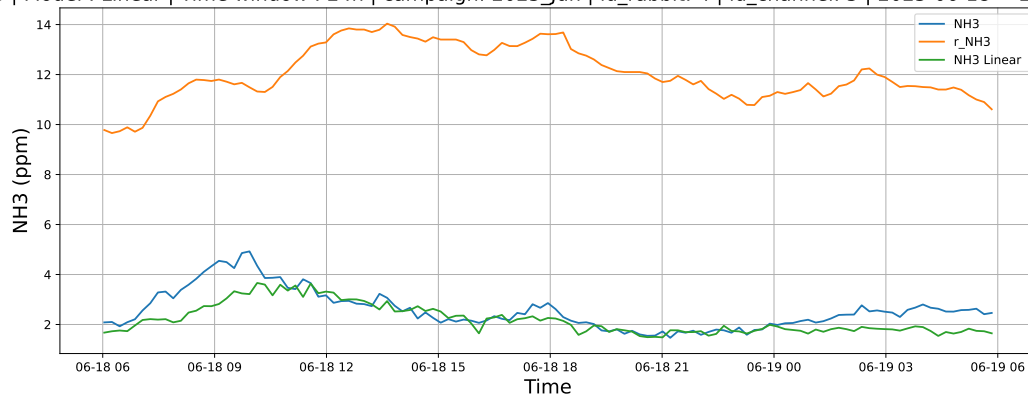
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



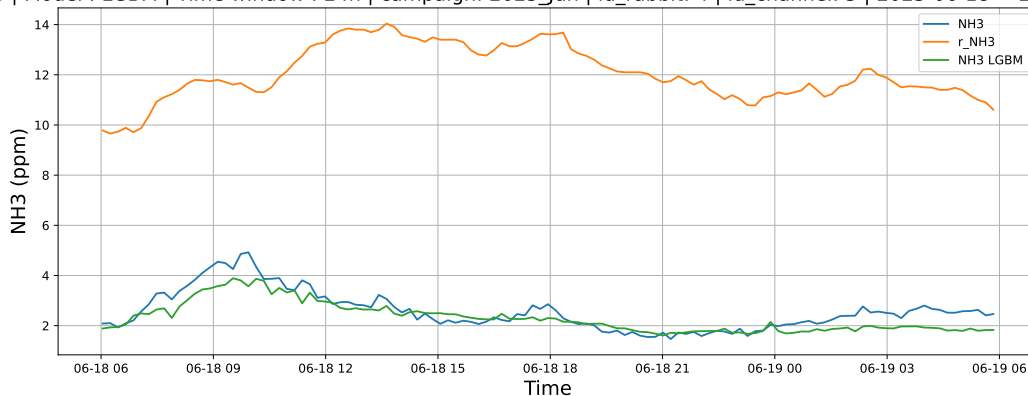
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

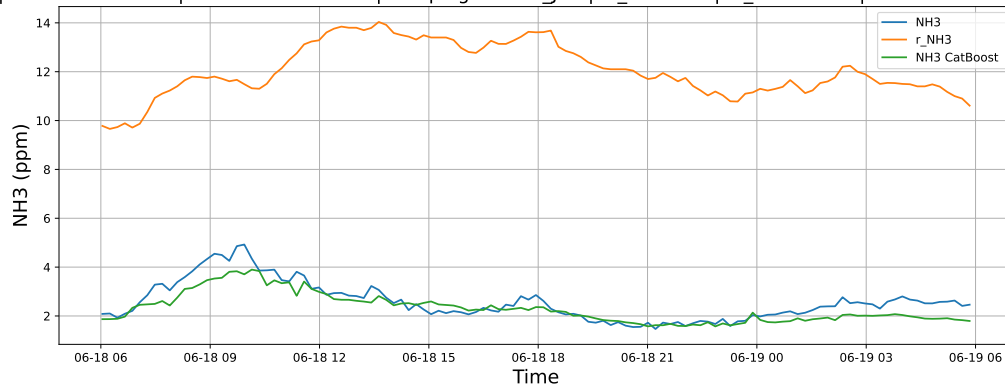
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

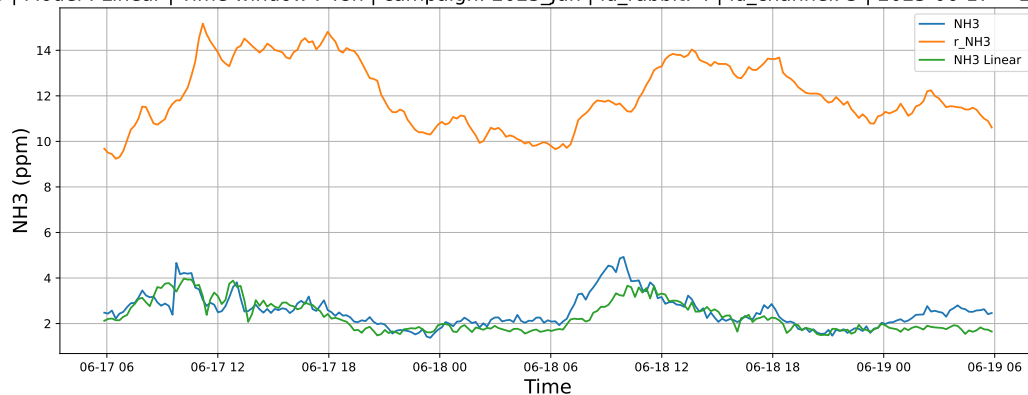


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

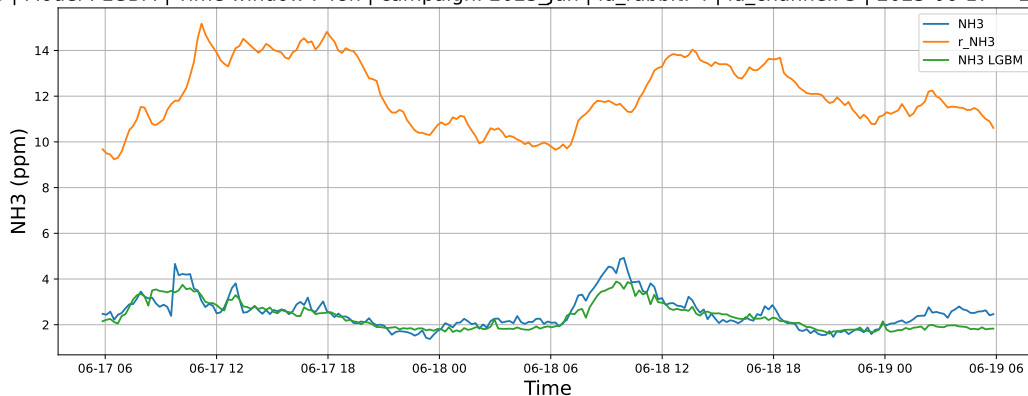


4.1.3.2 Time window : 48

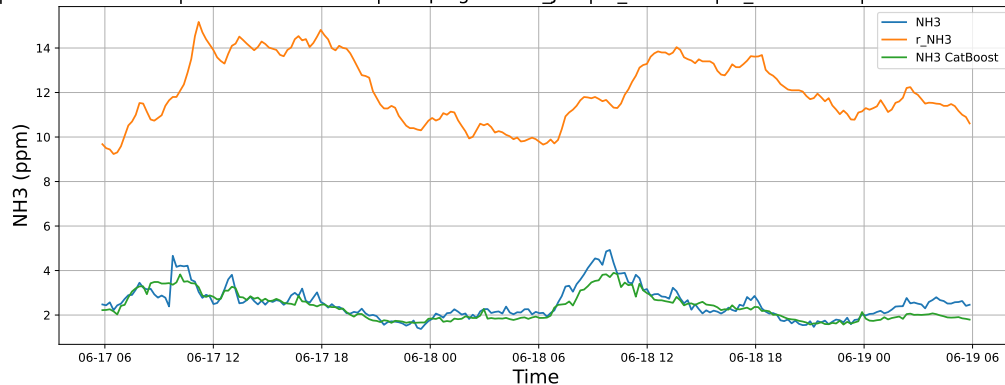
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

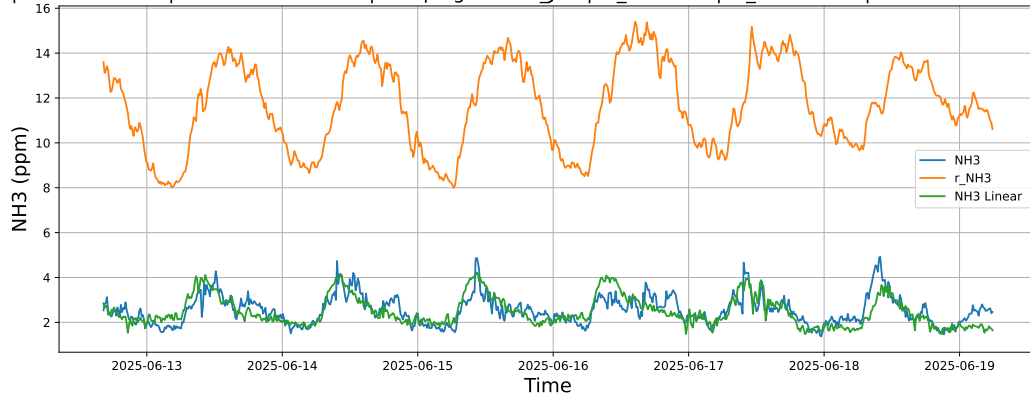


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

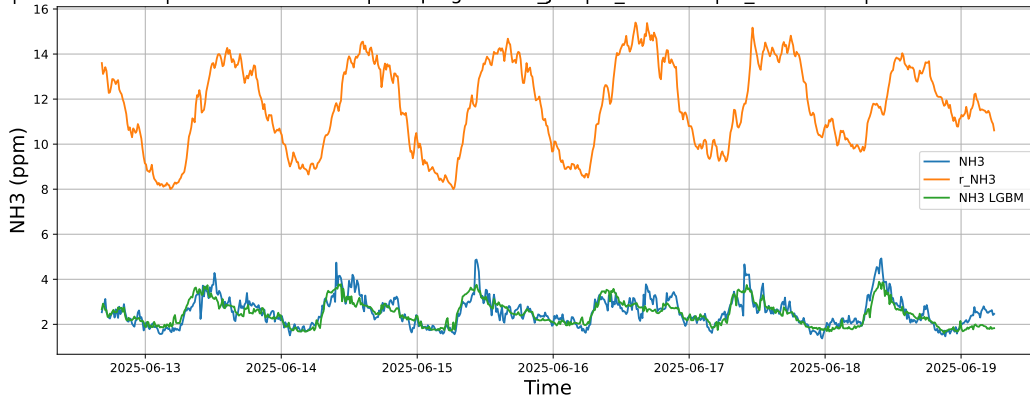


4.1.3.3 Time window : ALL

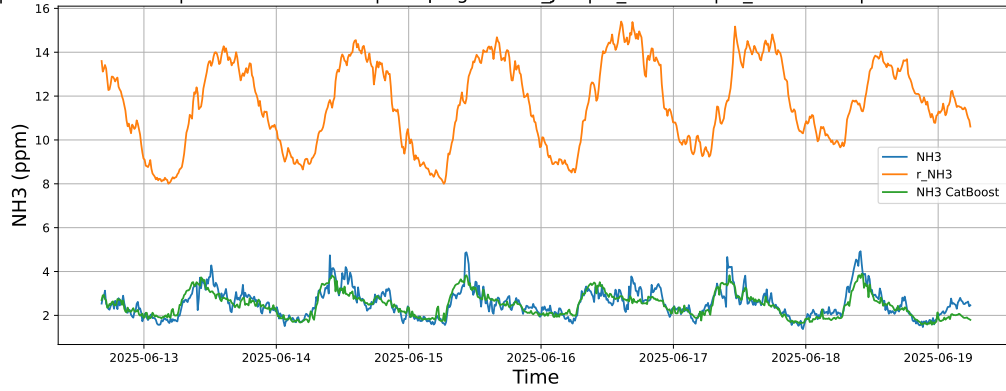
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



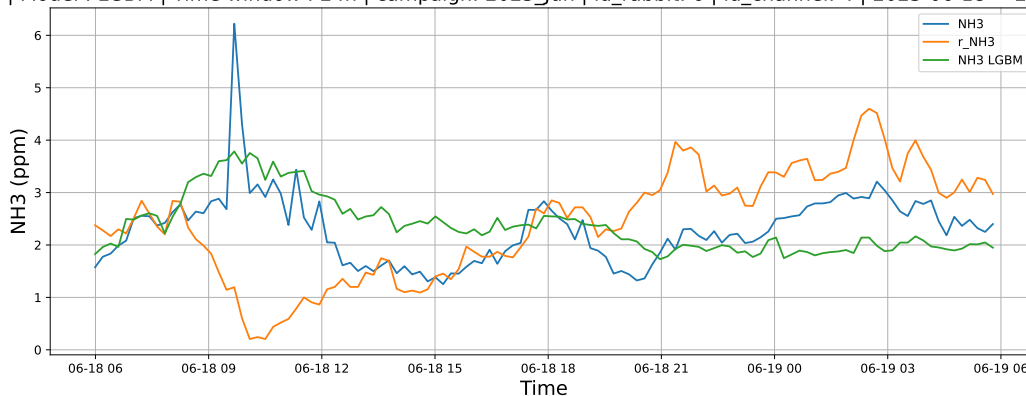
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

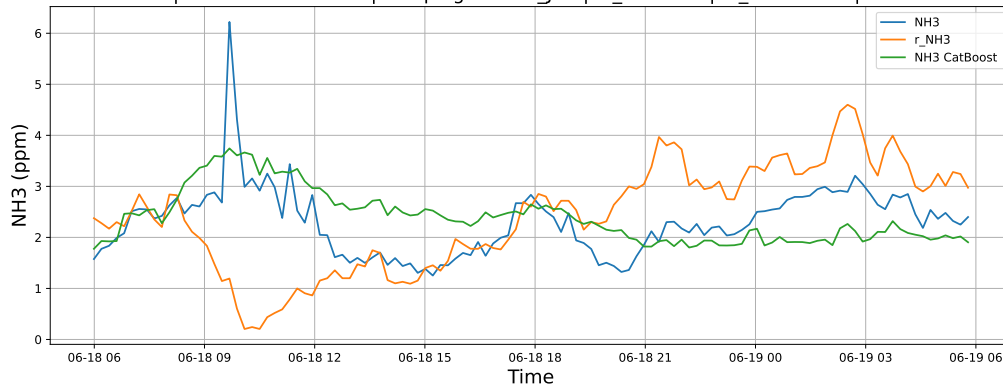
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

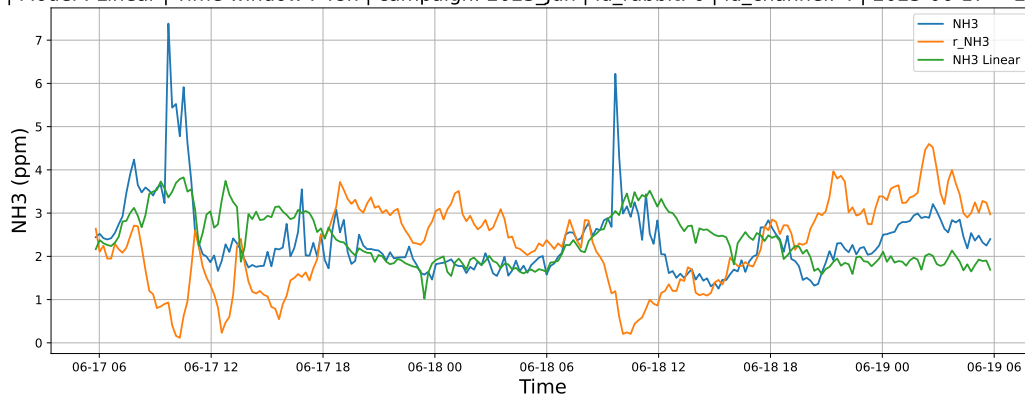


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

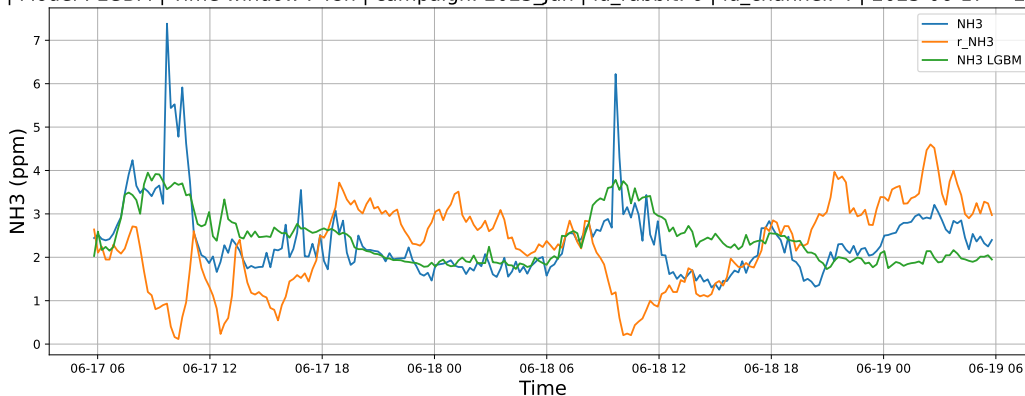


4.1.4.2 Time window : 48

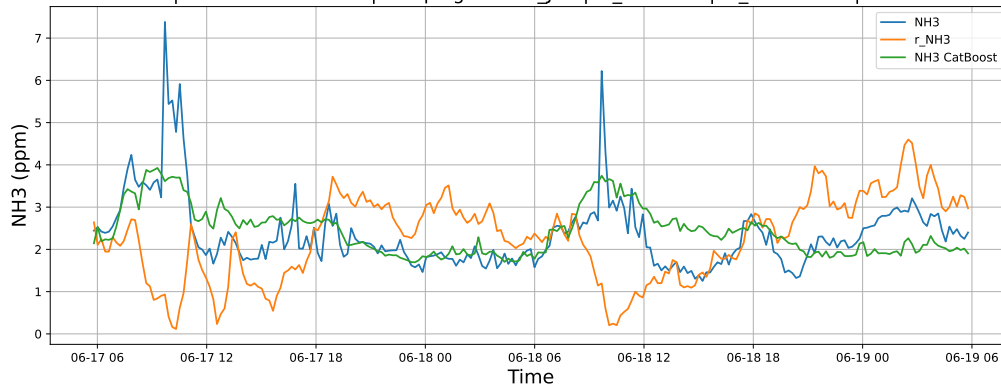
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

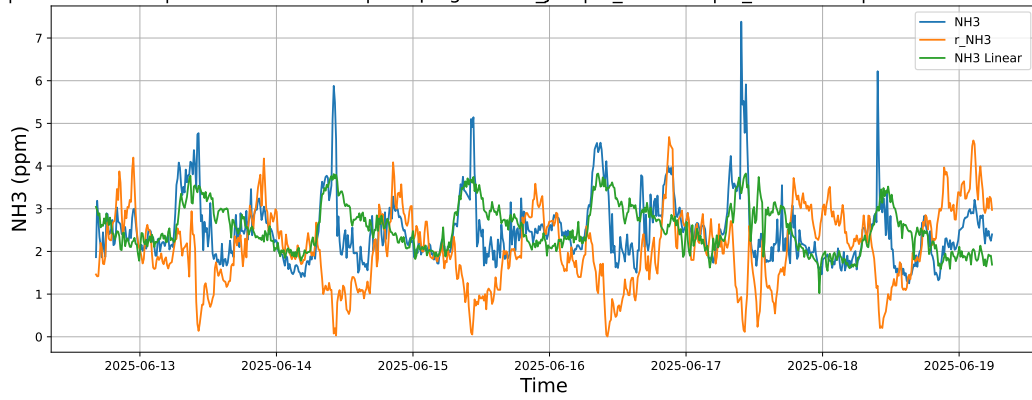


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

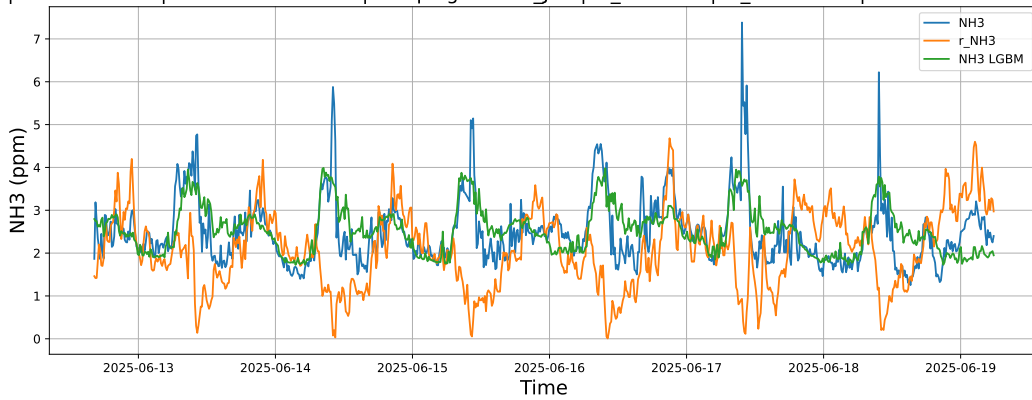


4.1.4.3 Time window : ALL

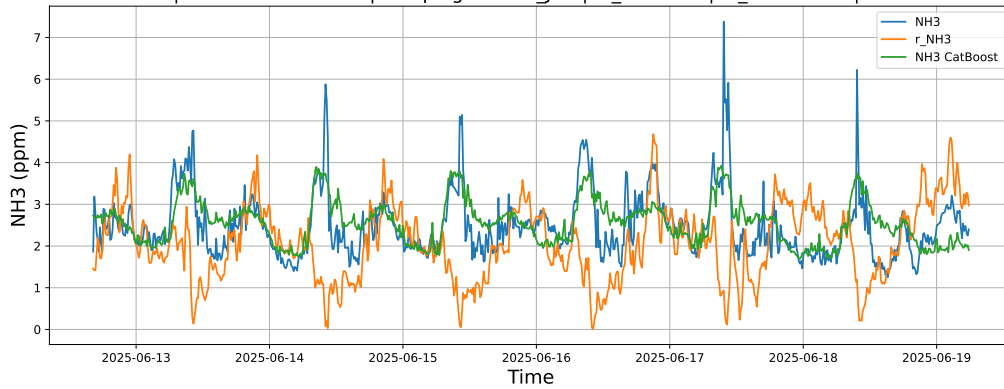
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



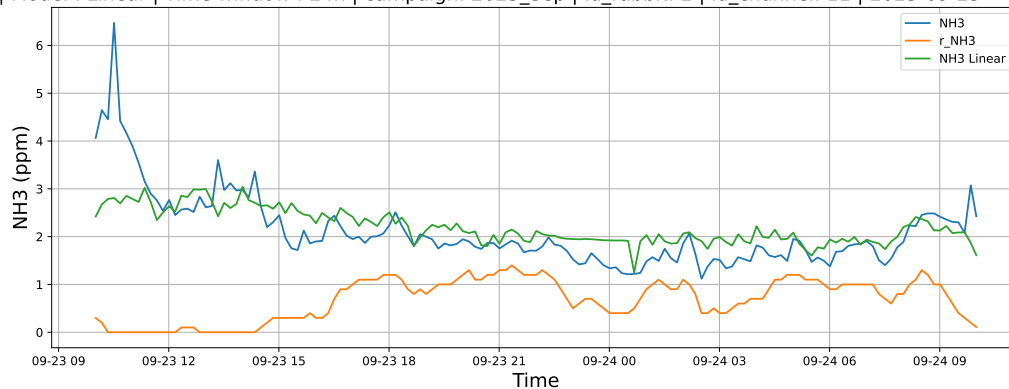
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



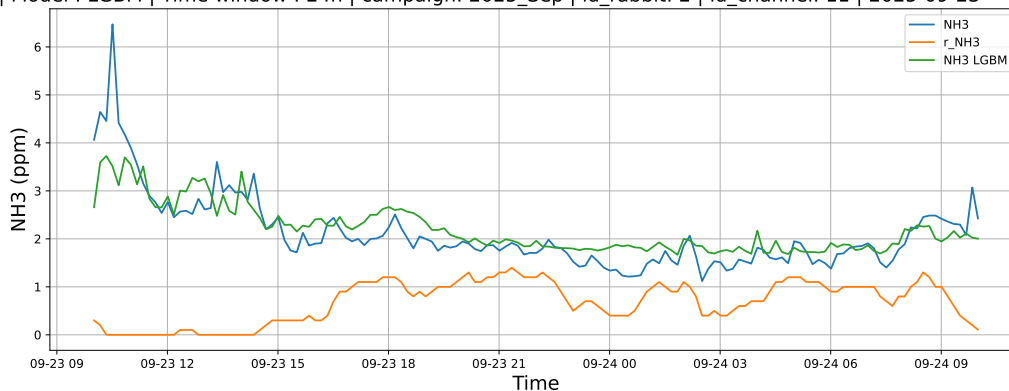
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

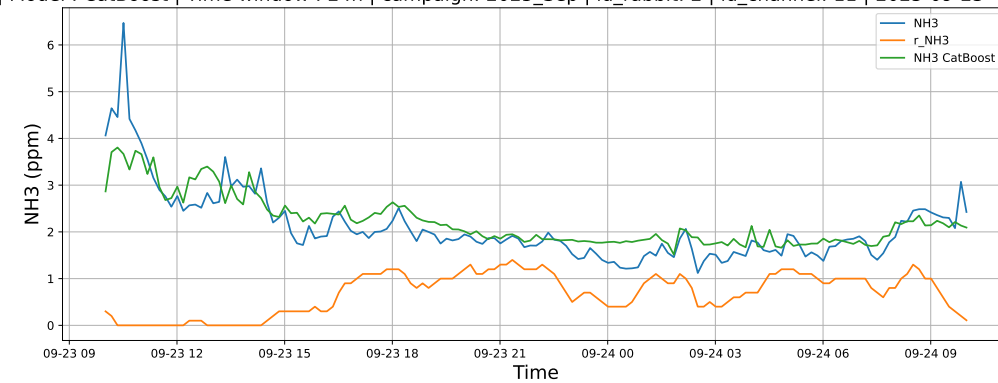
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

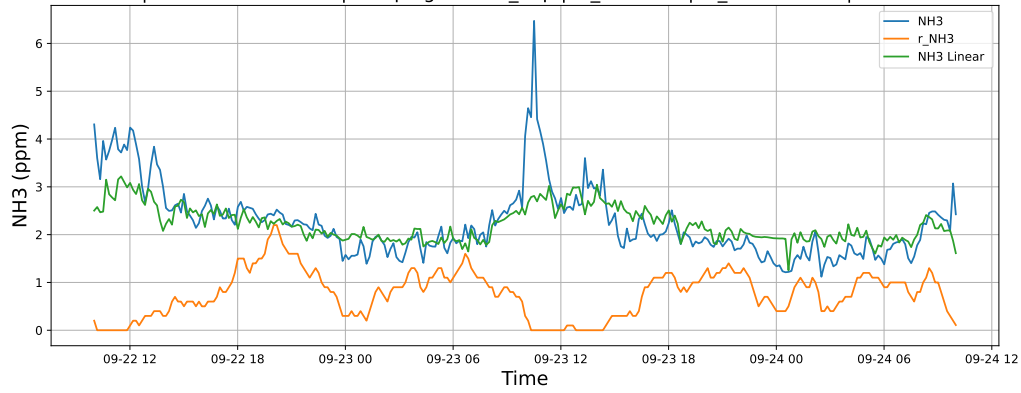


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

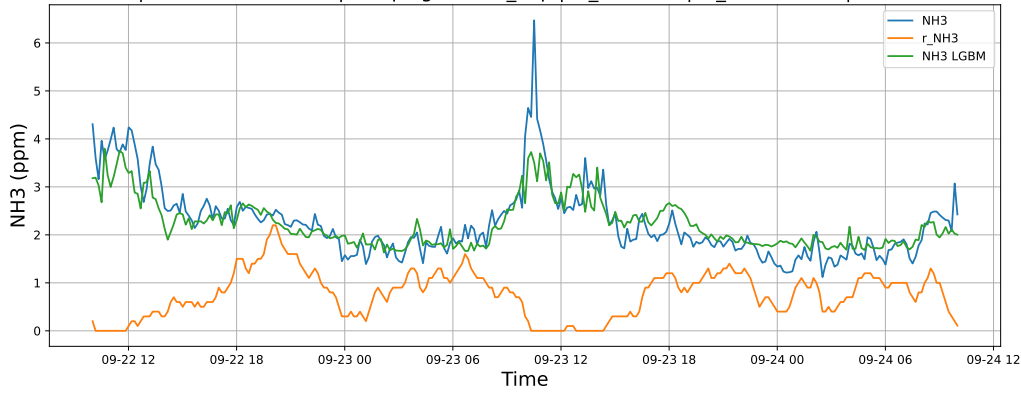


4.1.5.2 Time window : 48

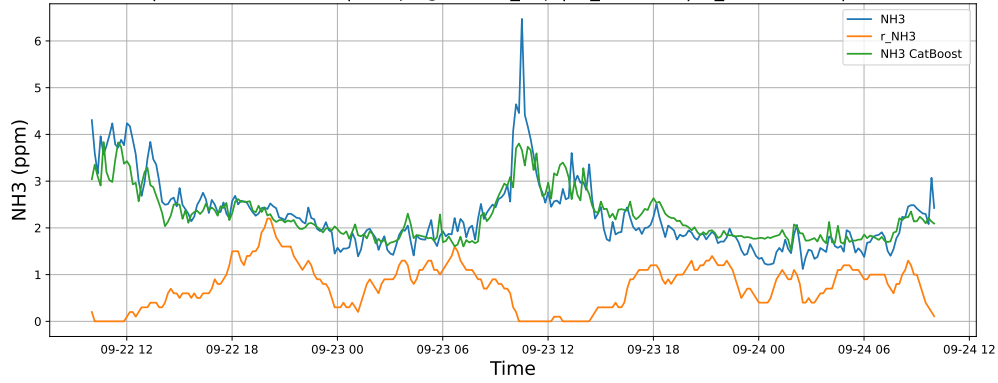
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

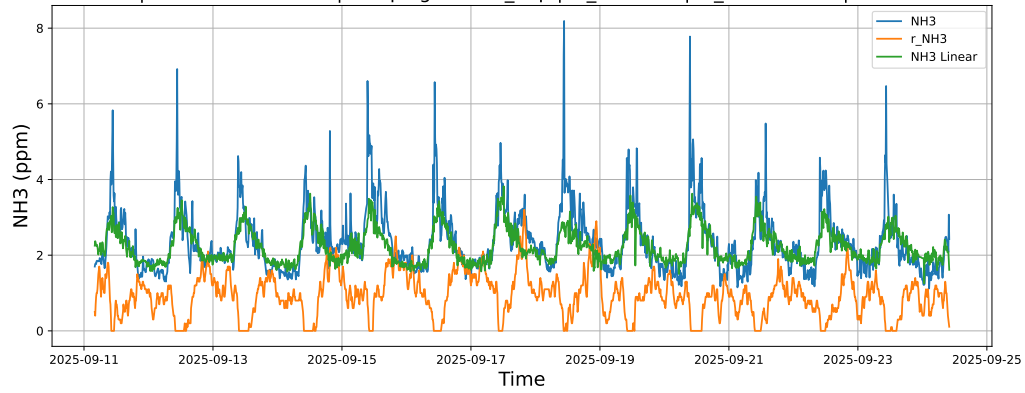


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

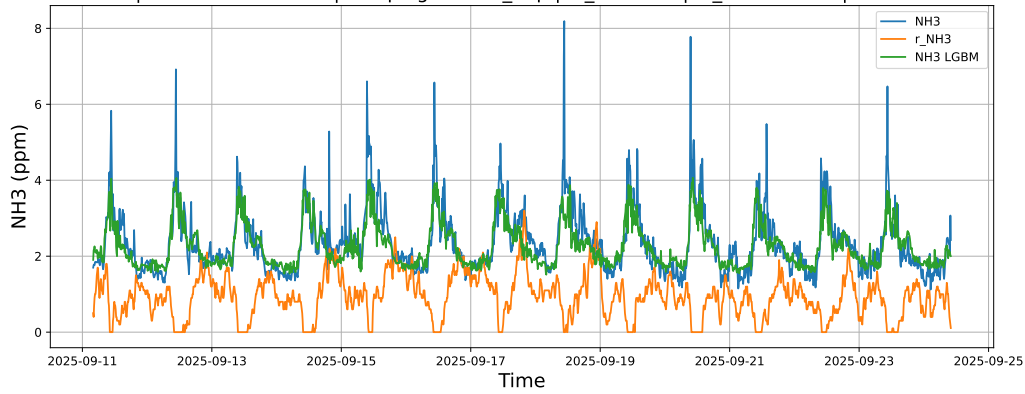


4.1.5.3 Time window : ALL

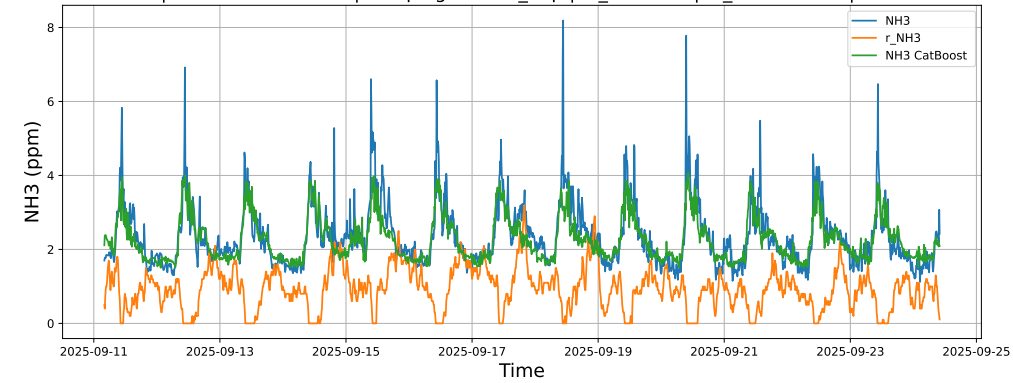
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



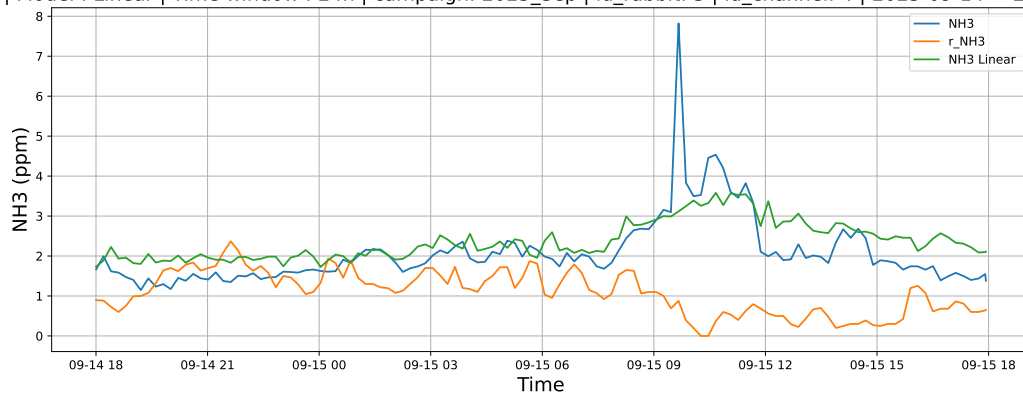
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



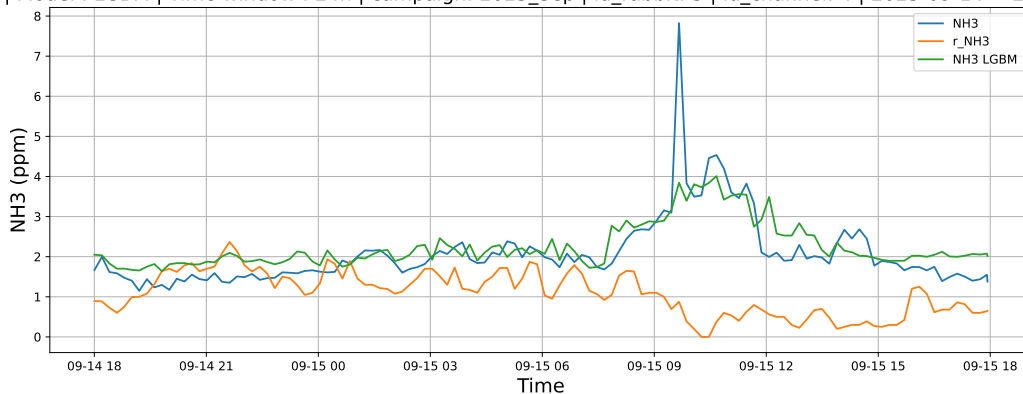
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

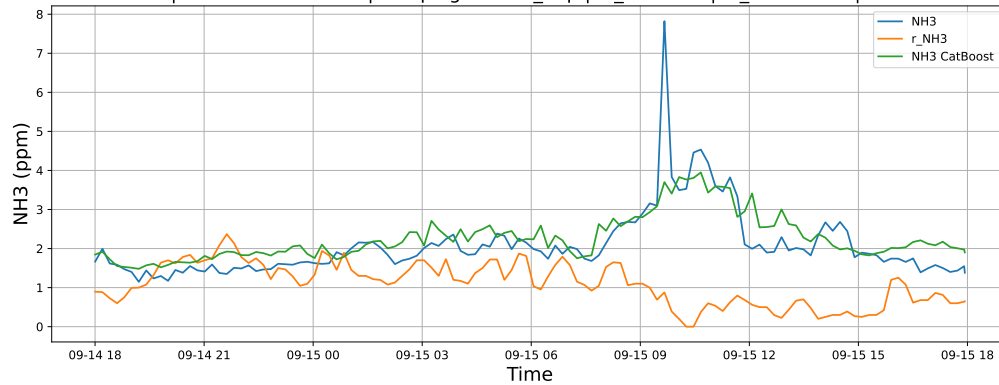
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

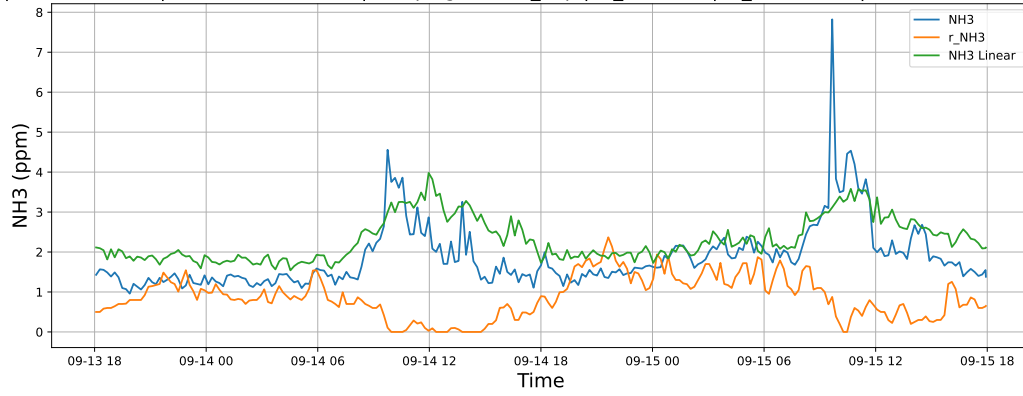


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

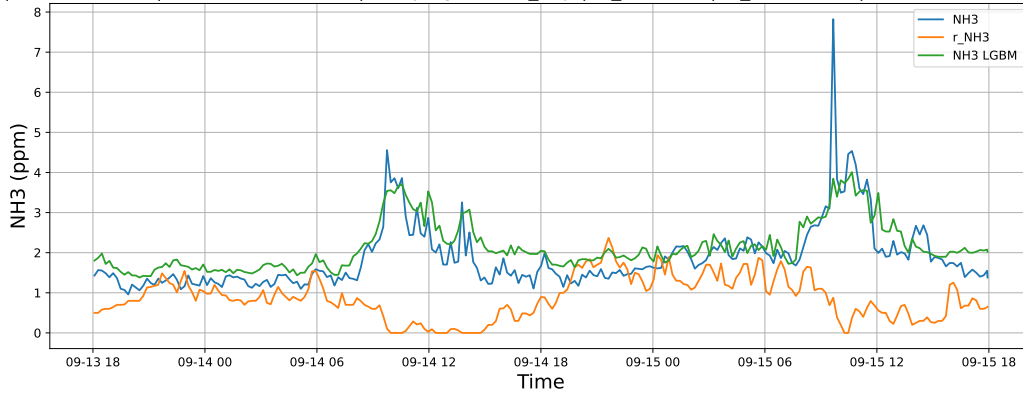


4.1.6.2 Time window : 48

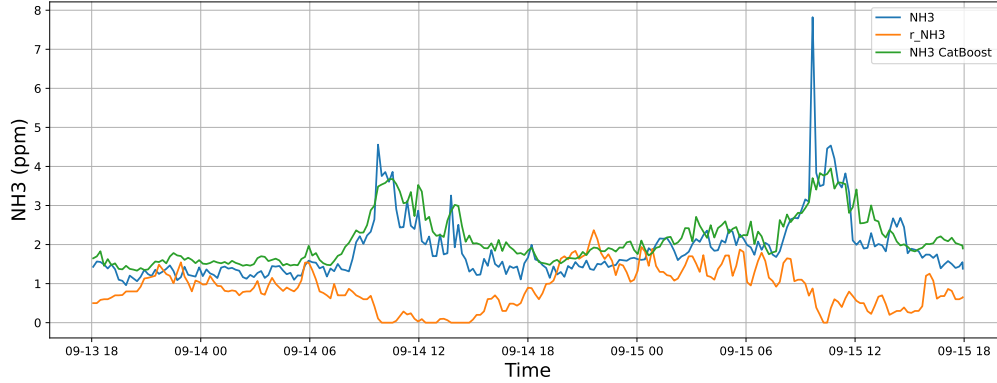
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

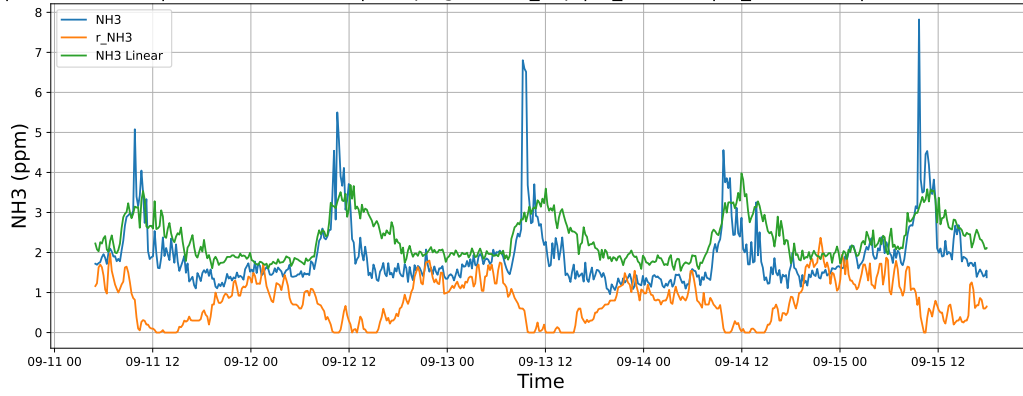


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

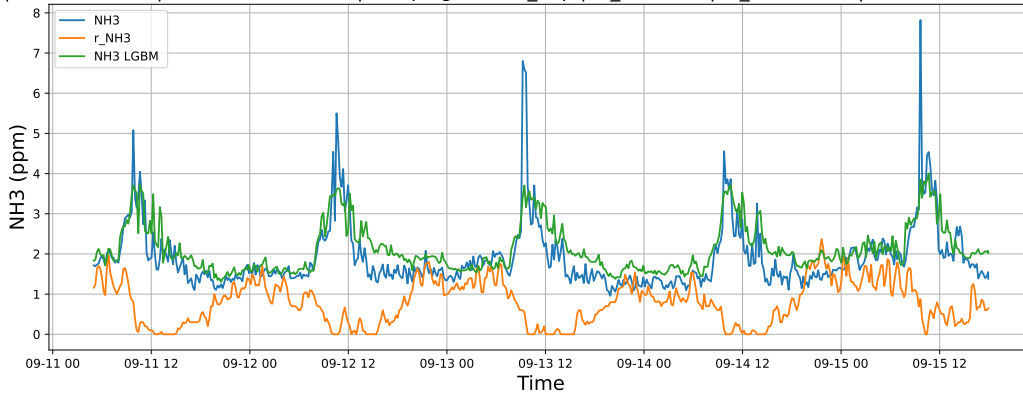


4.1.6.3 Time window : ALL

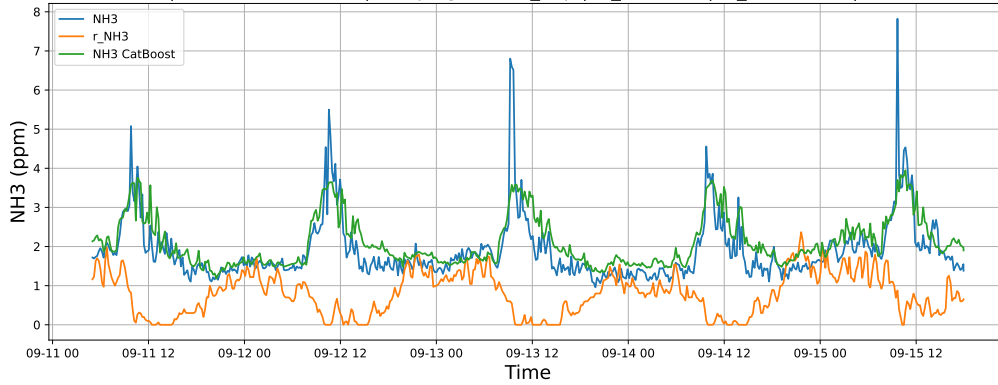
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



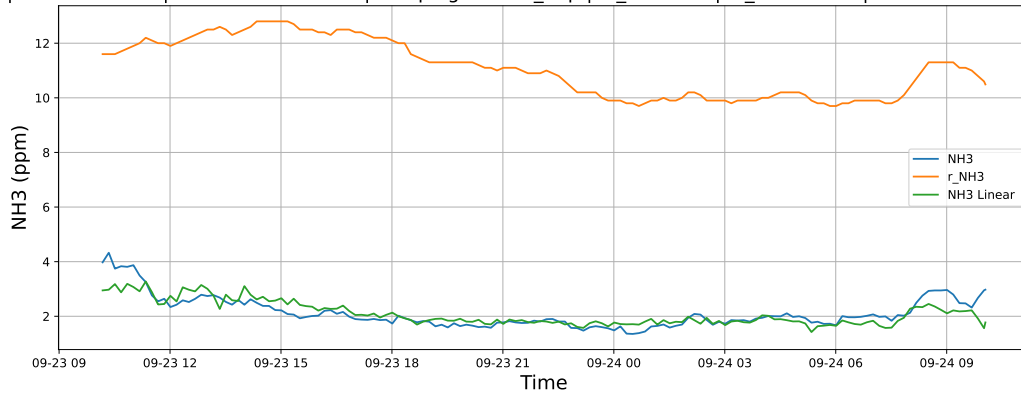
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



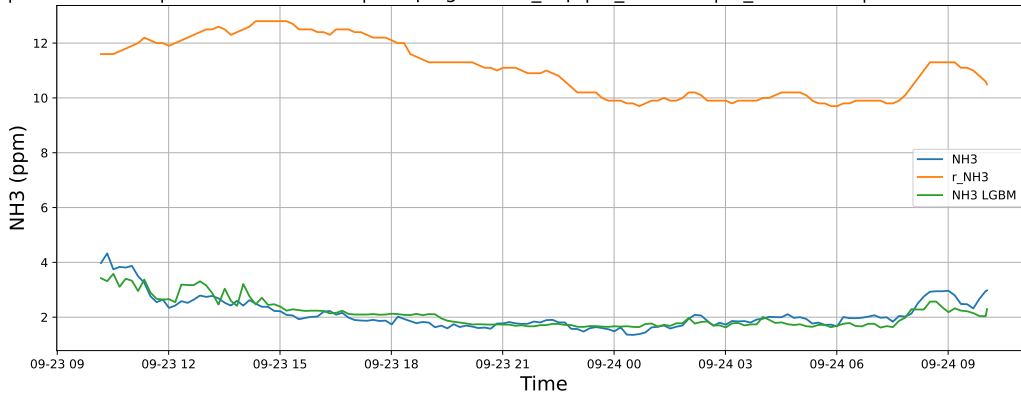
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

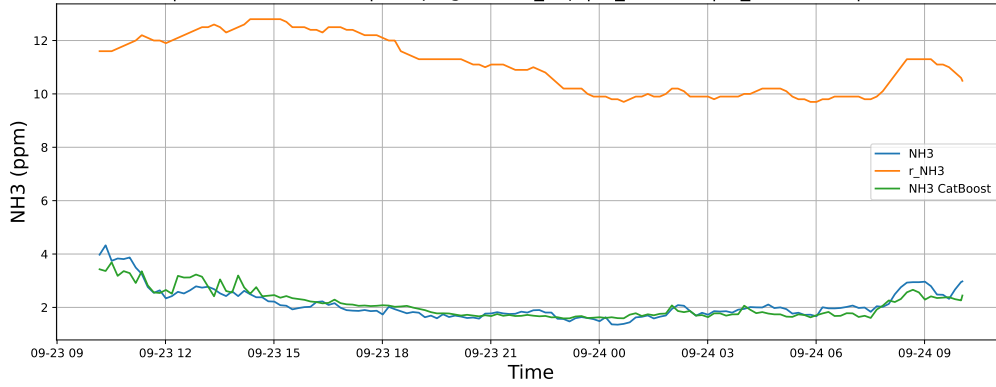
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

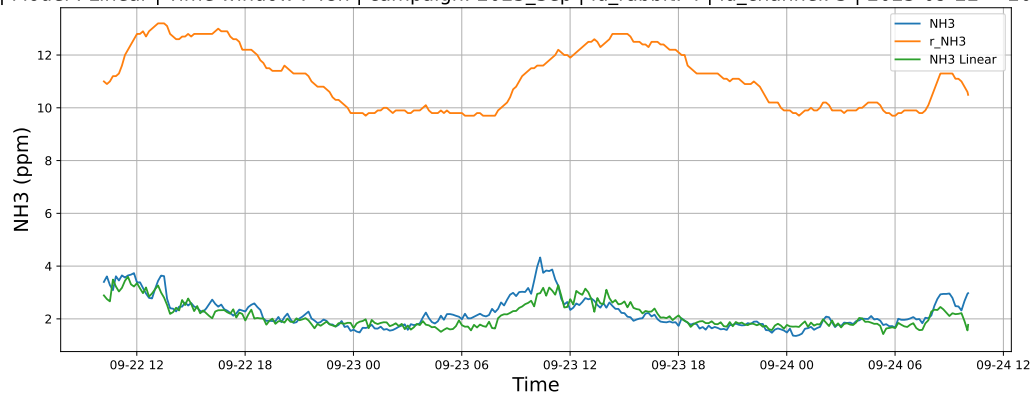


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

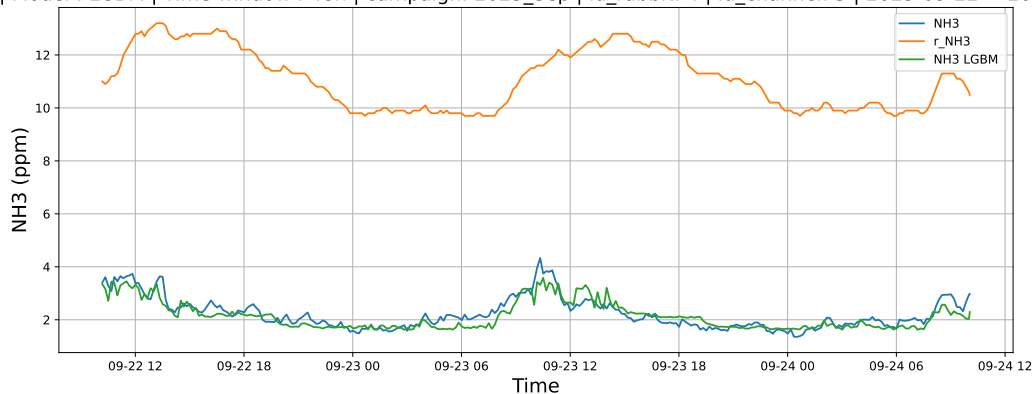


4.1.7.2 Time window : 48

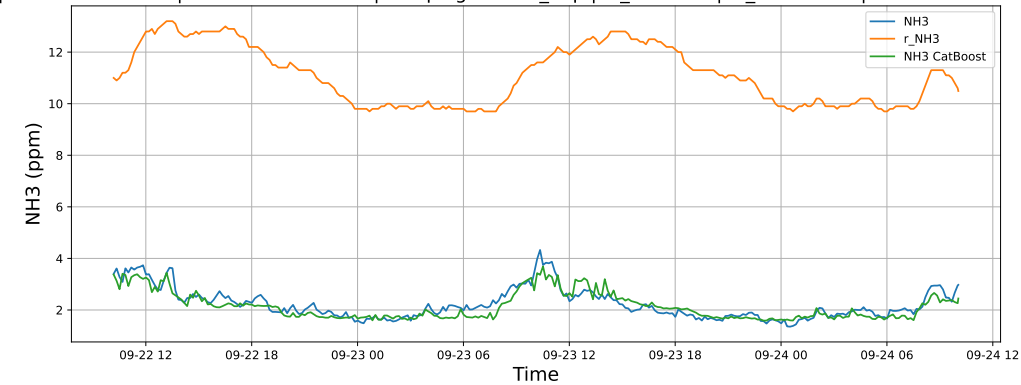
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

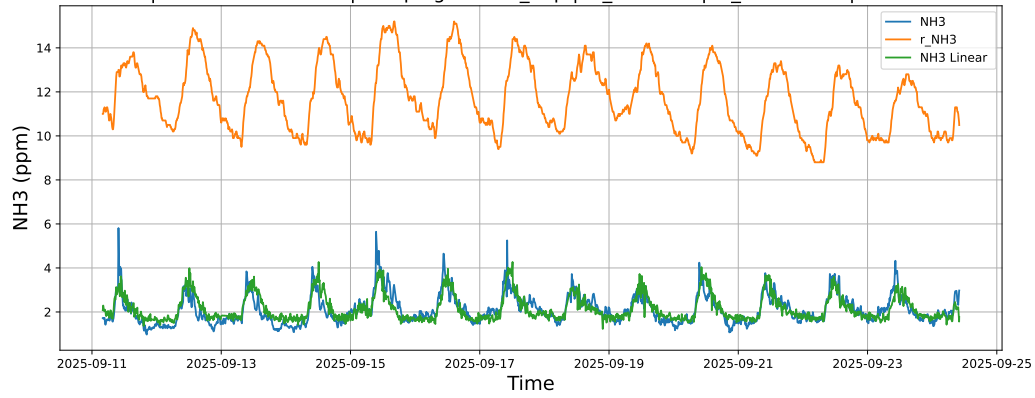


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

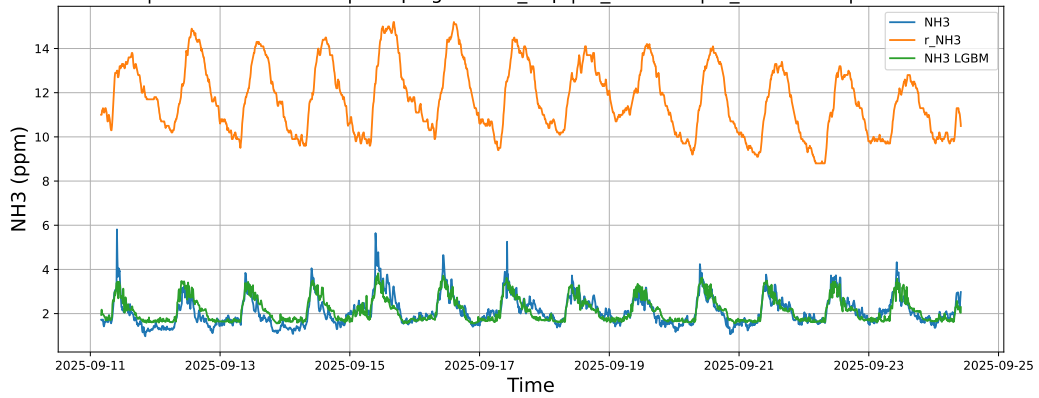


4.1.7.3 Time window : ALL

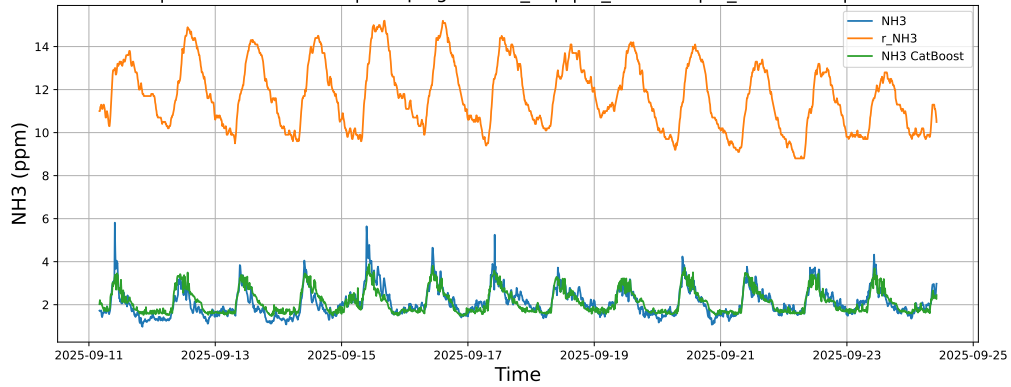
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



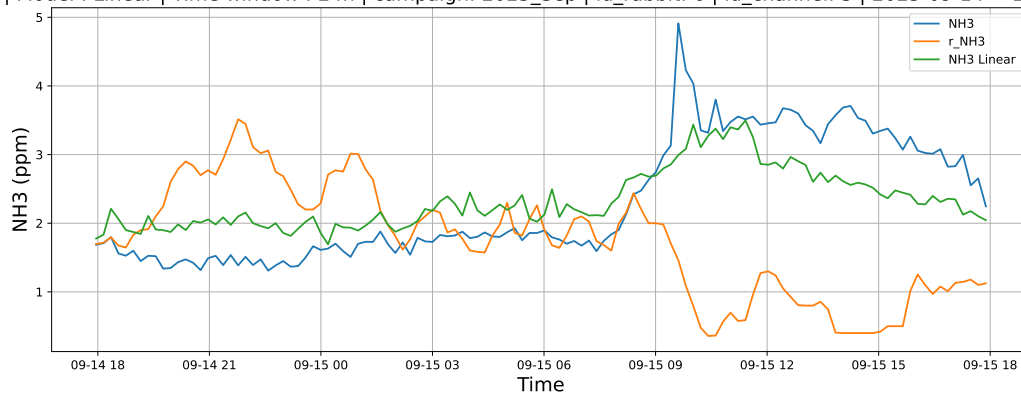
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



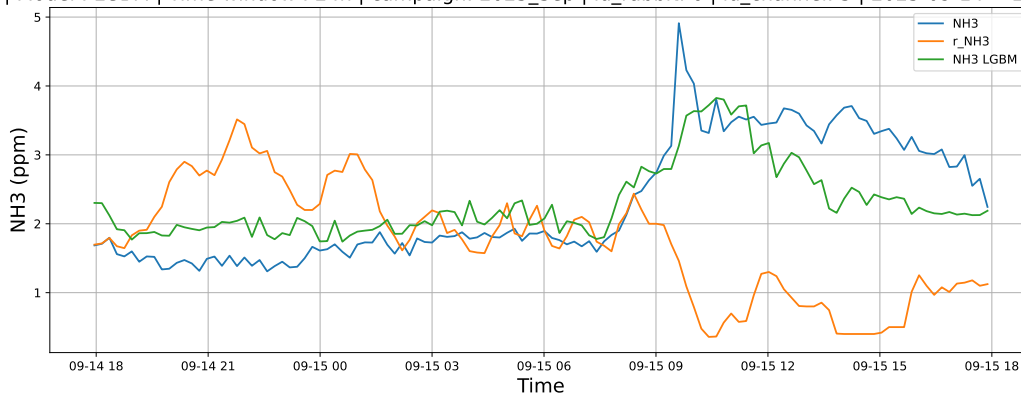
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

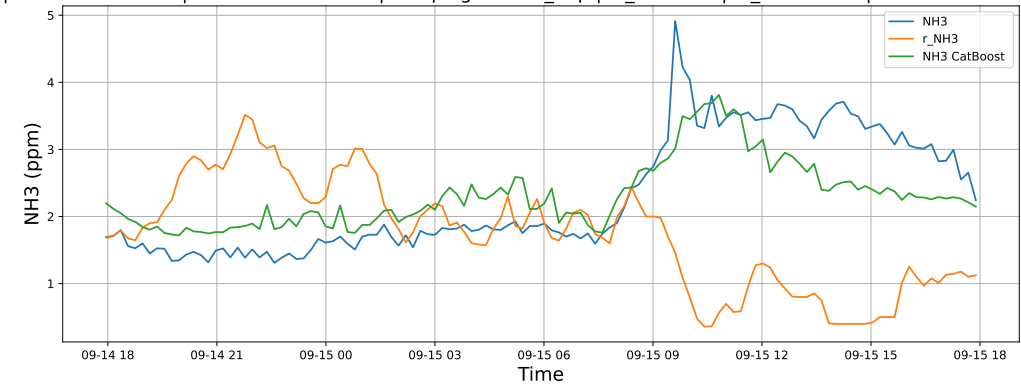
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

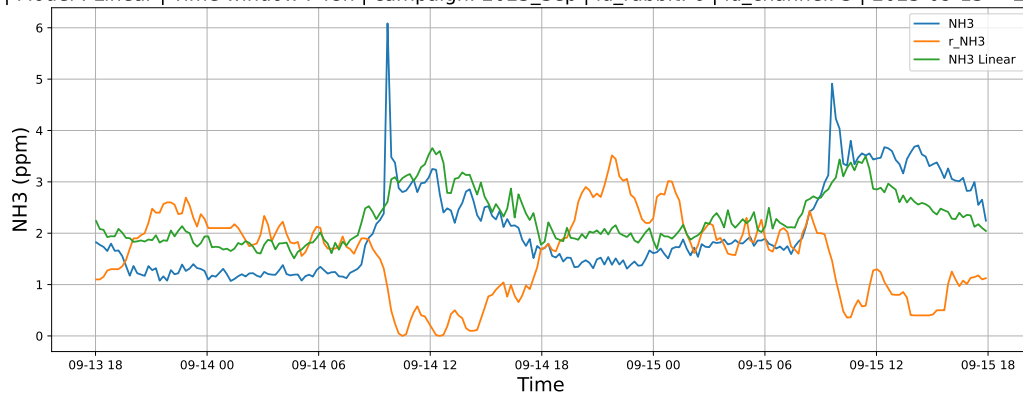


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

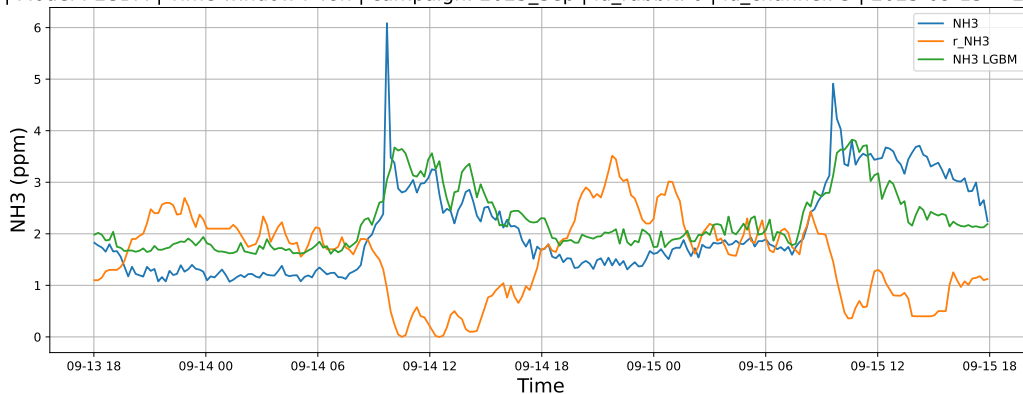


4.1.8.2 Time window : 48

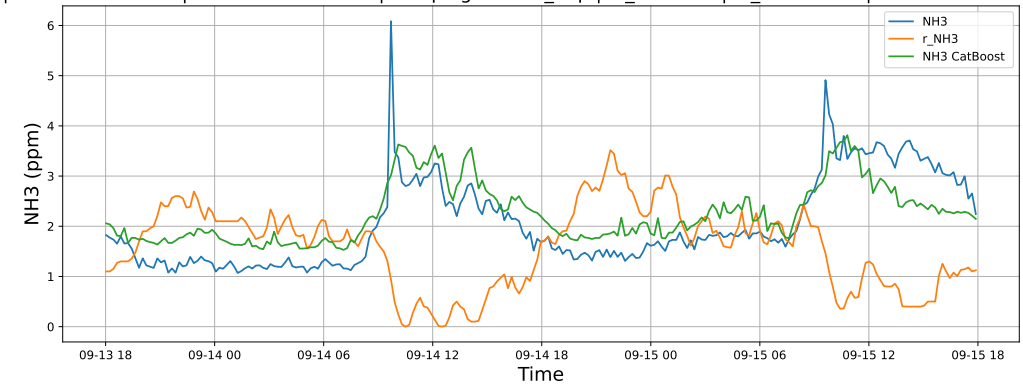
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

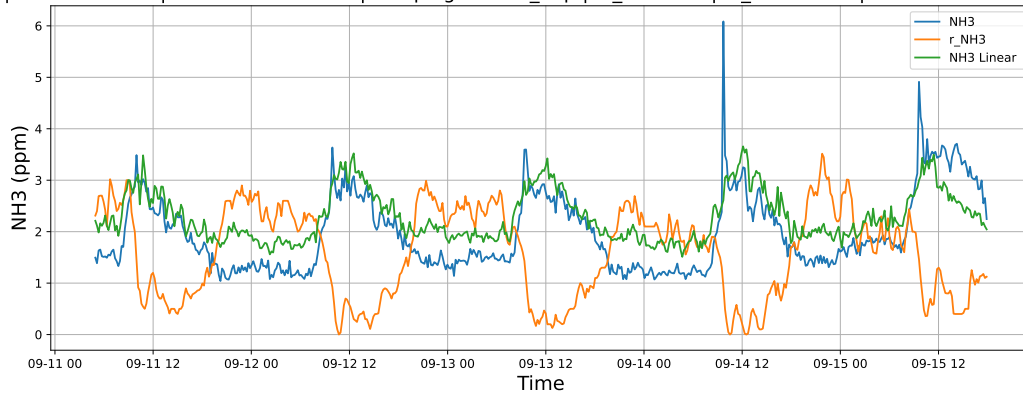


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

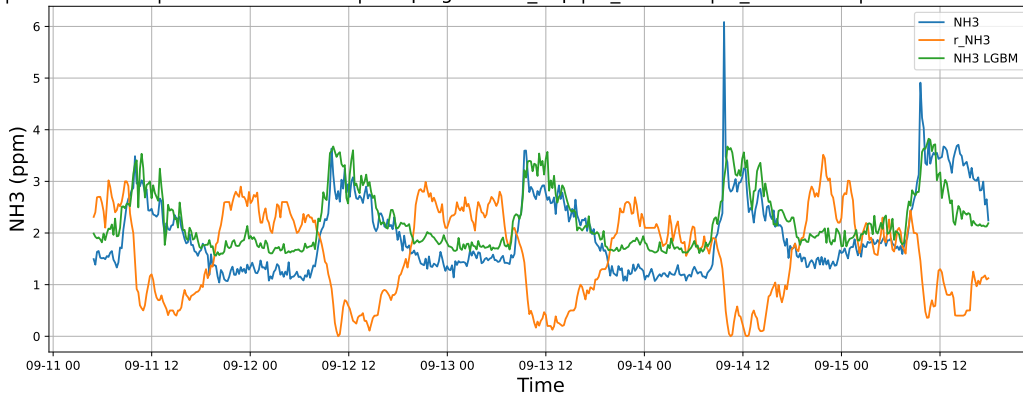


4.1.8.3 Time window : ALL

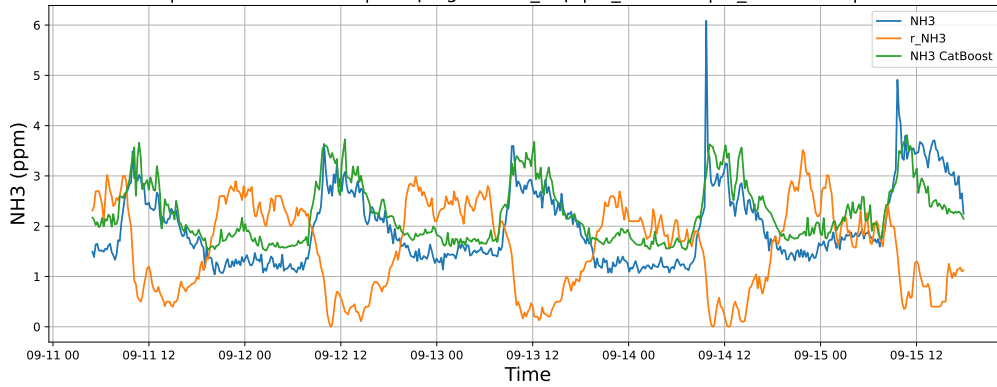
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

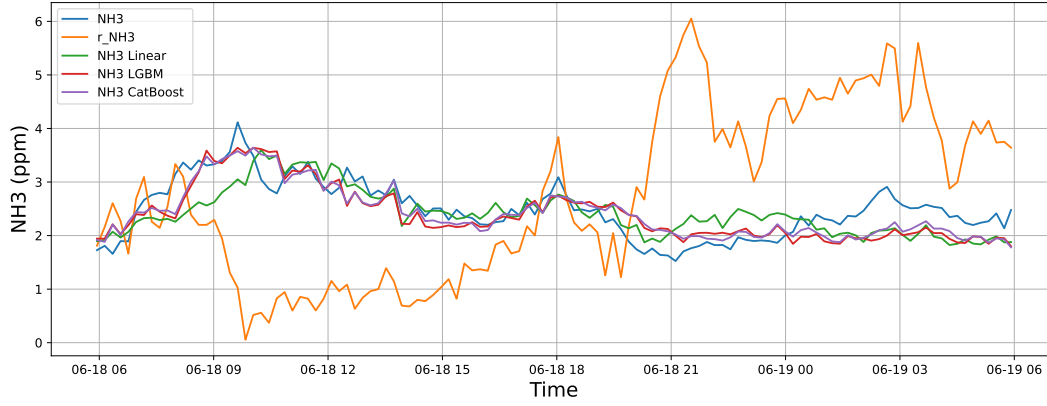


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

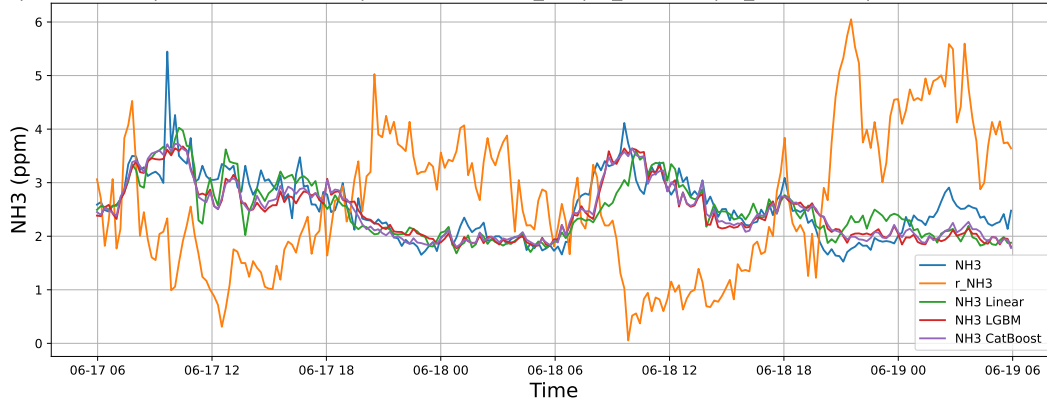
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



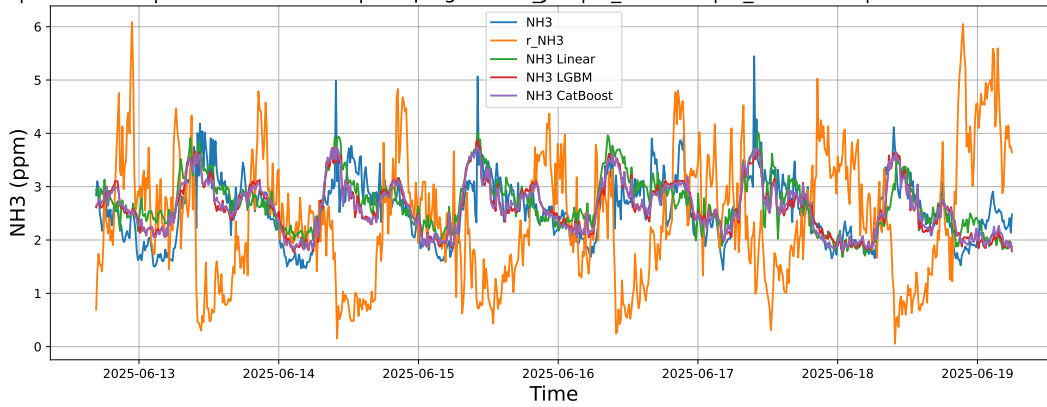
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

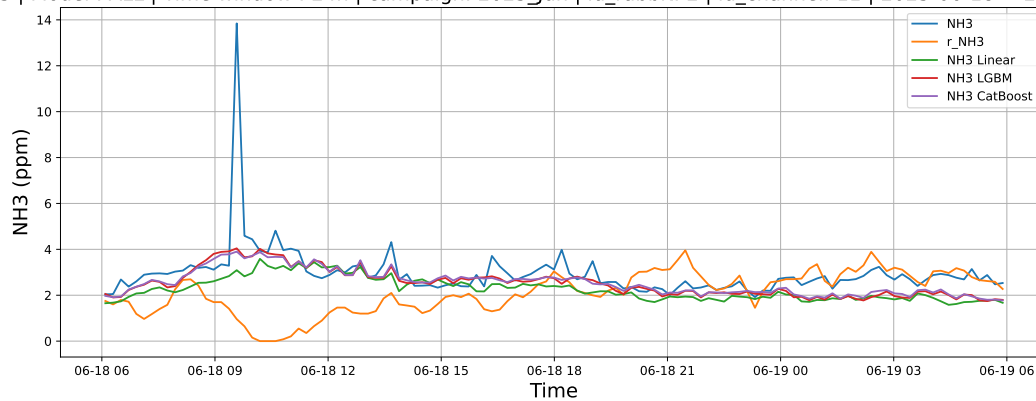
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

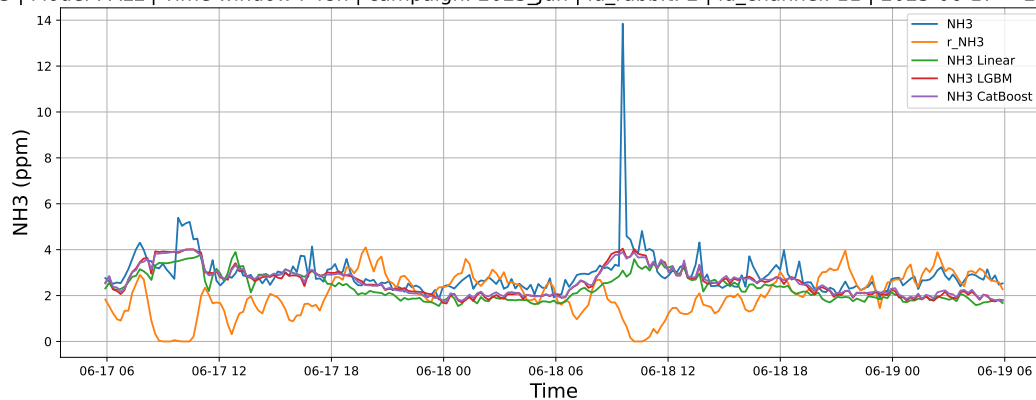
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



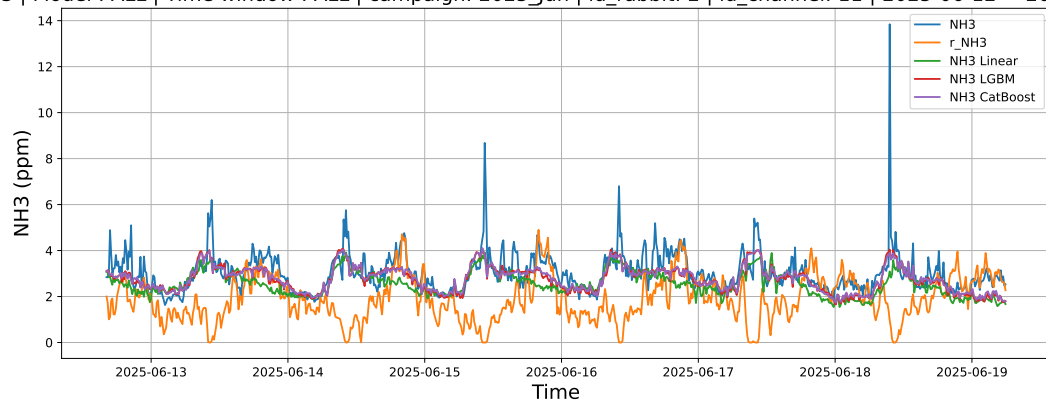
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

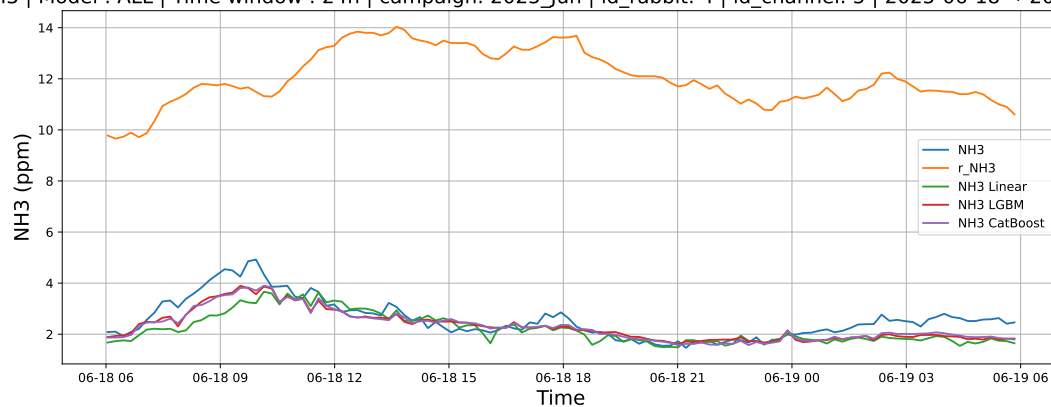
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

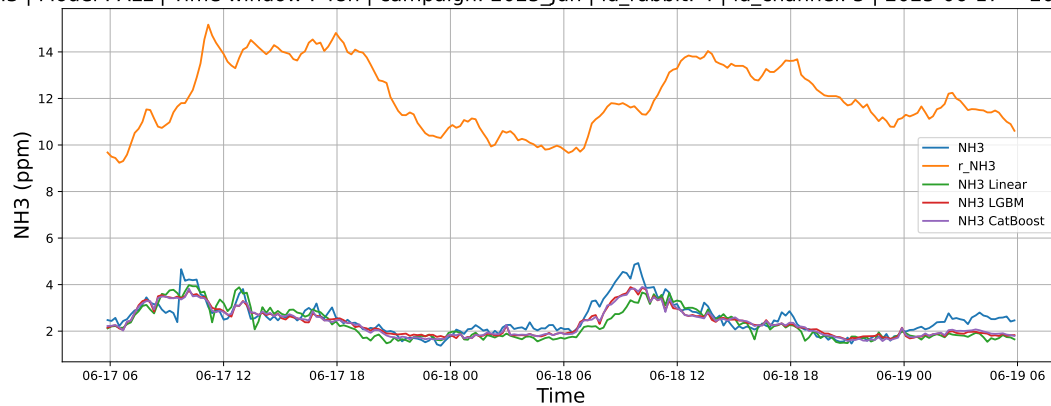
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



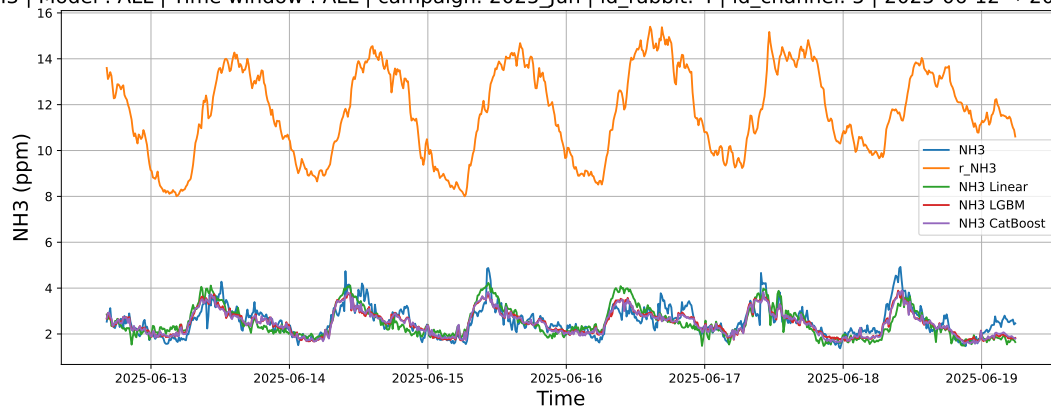
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

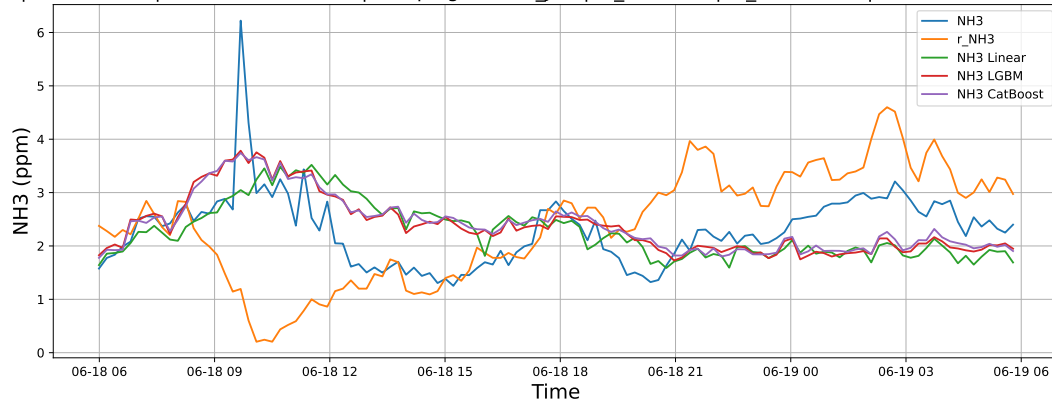
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

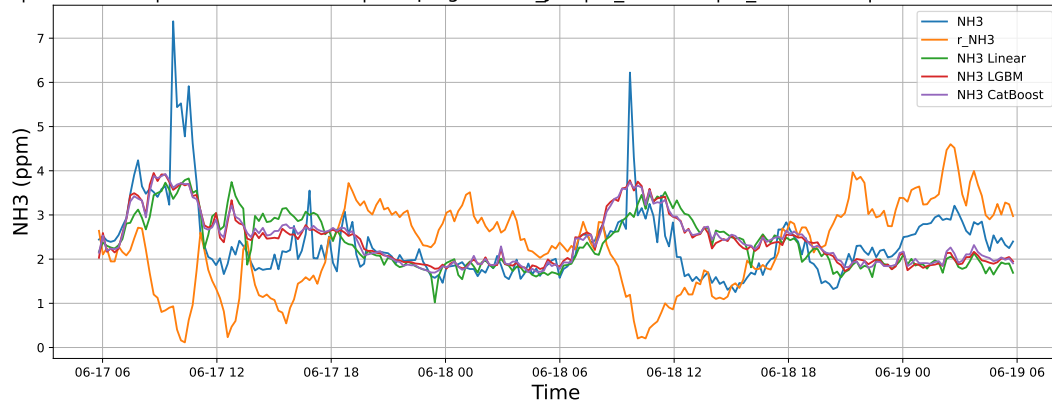
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



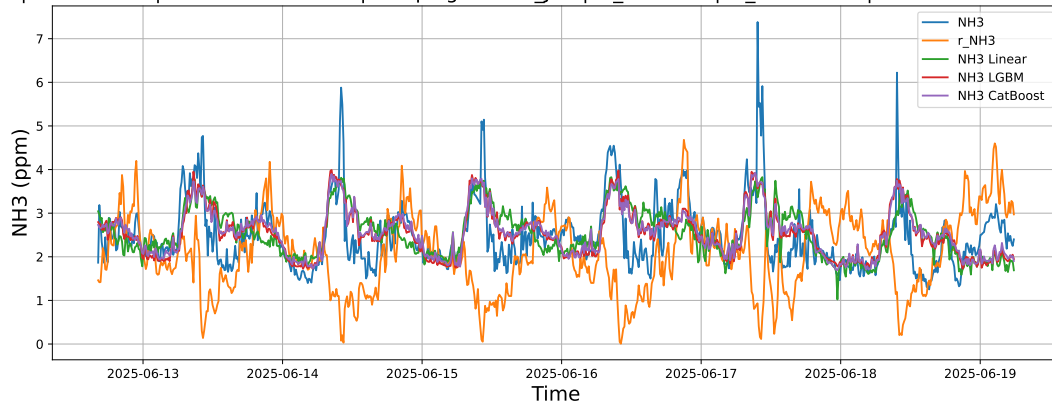
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

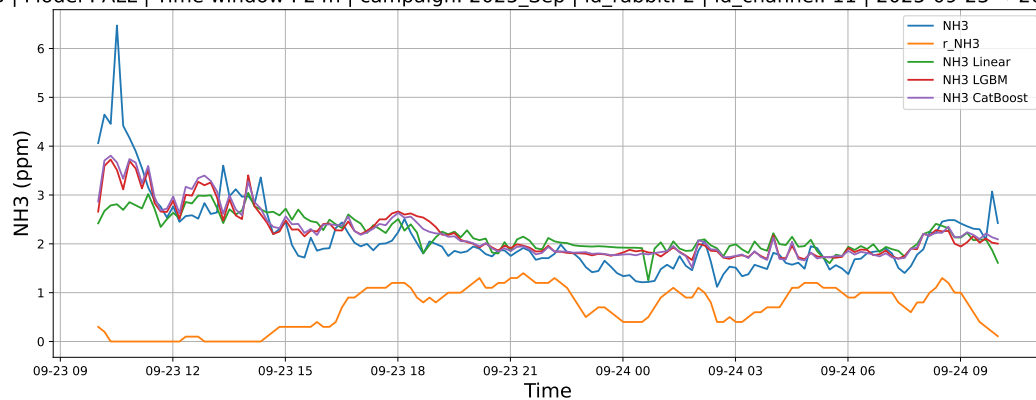
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

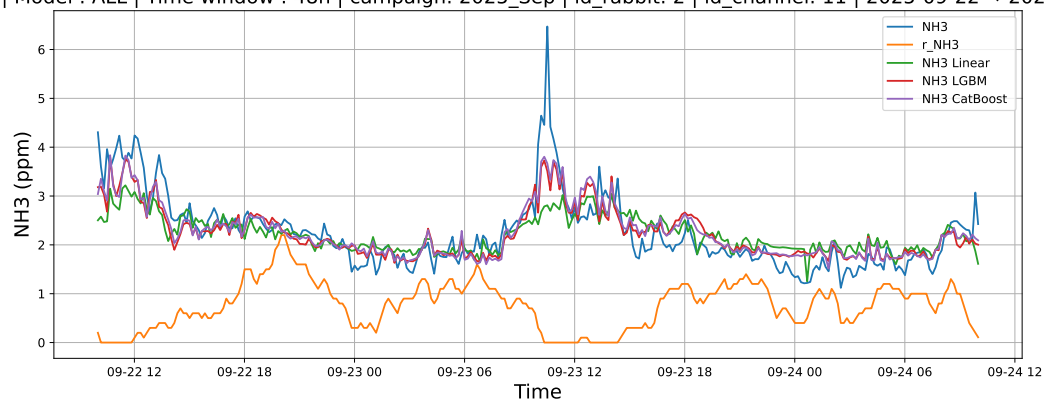
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



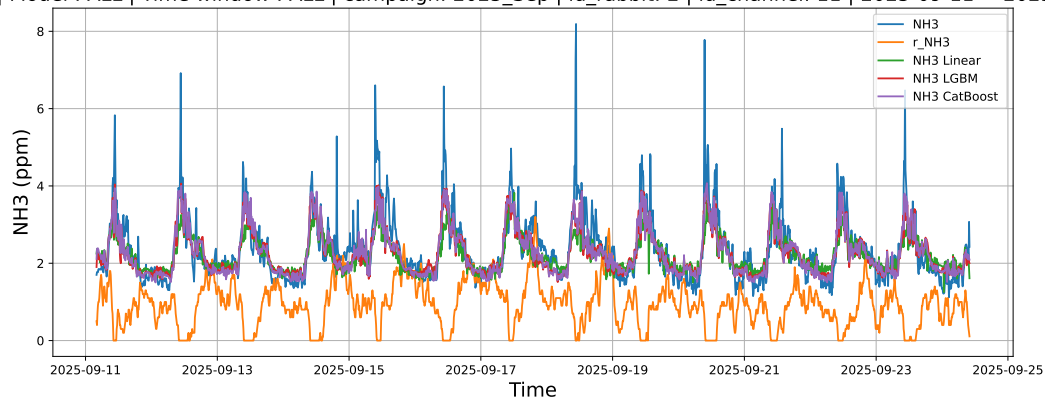
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

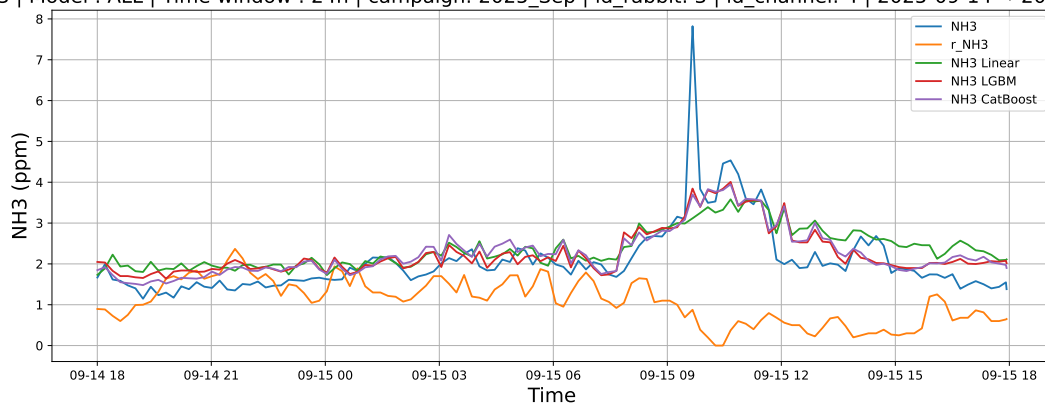
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

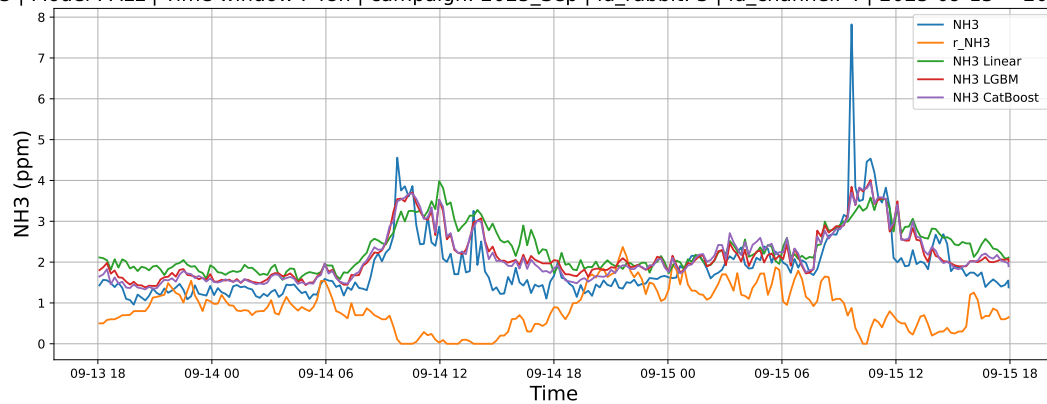
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



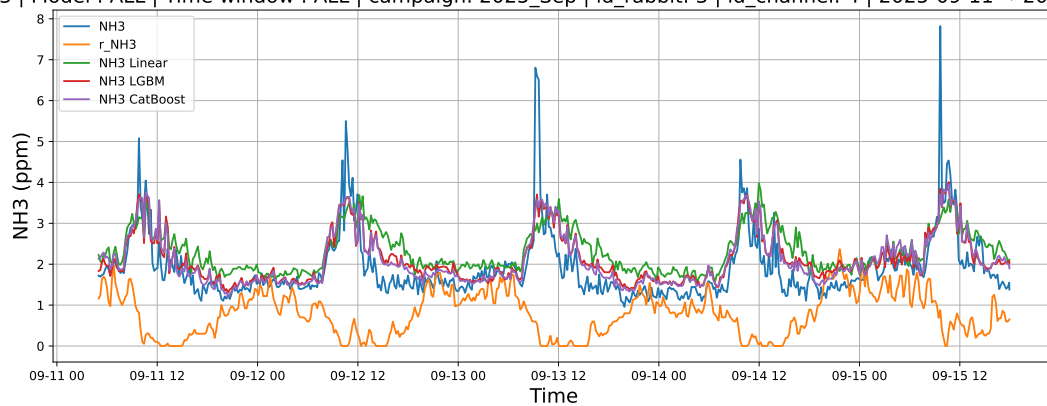
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

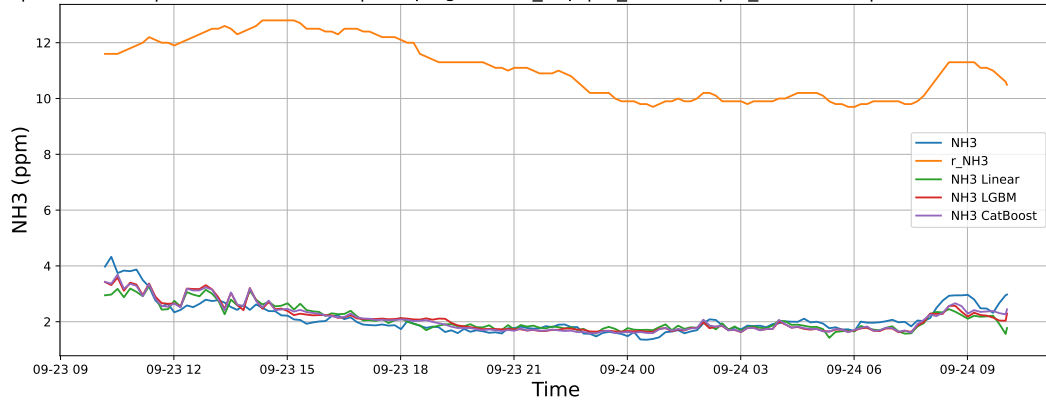
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

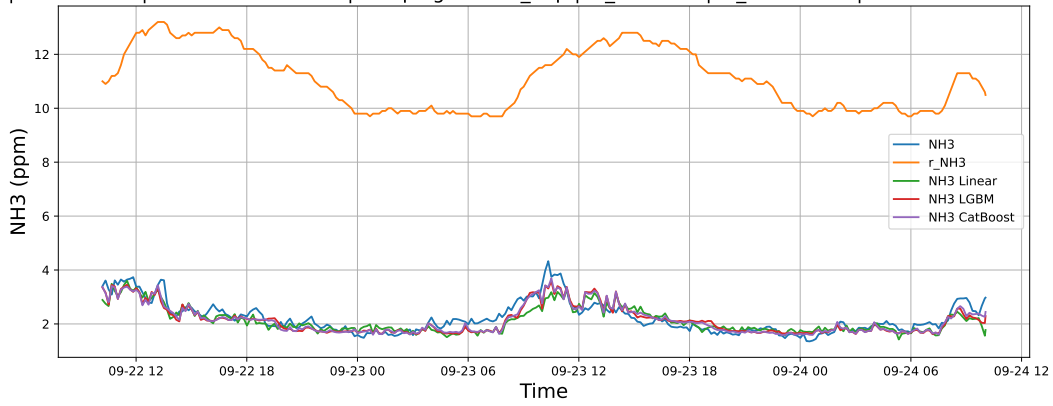
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



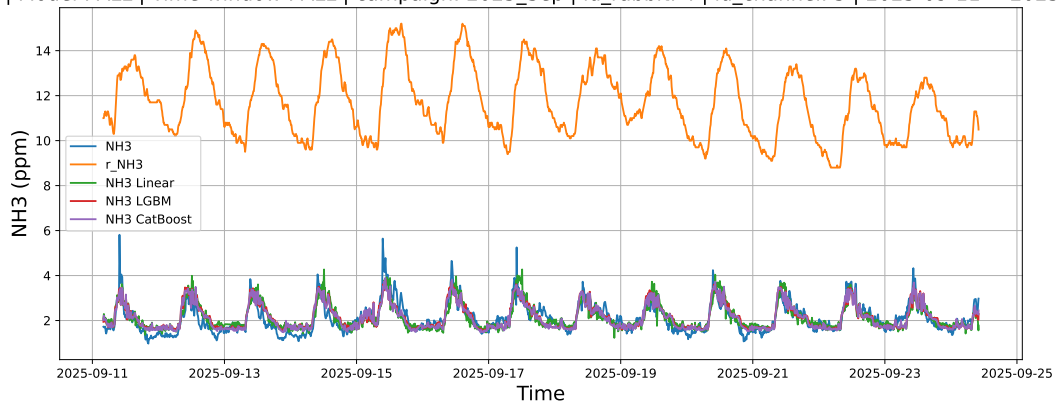
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

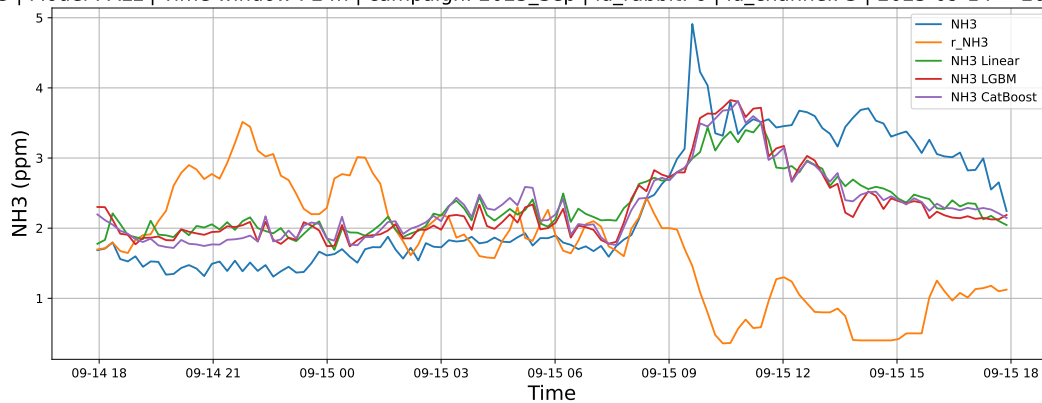
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

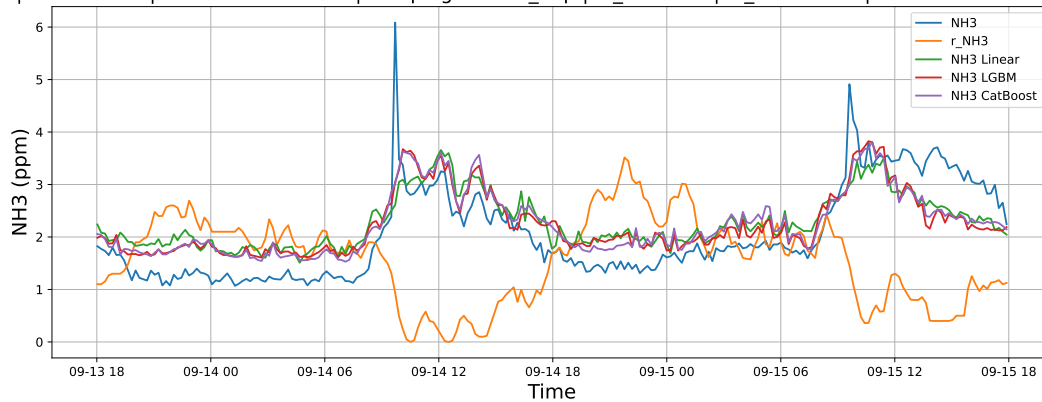
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

